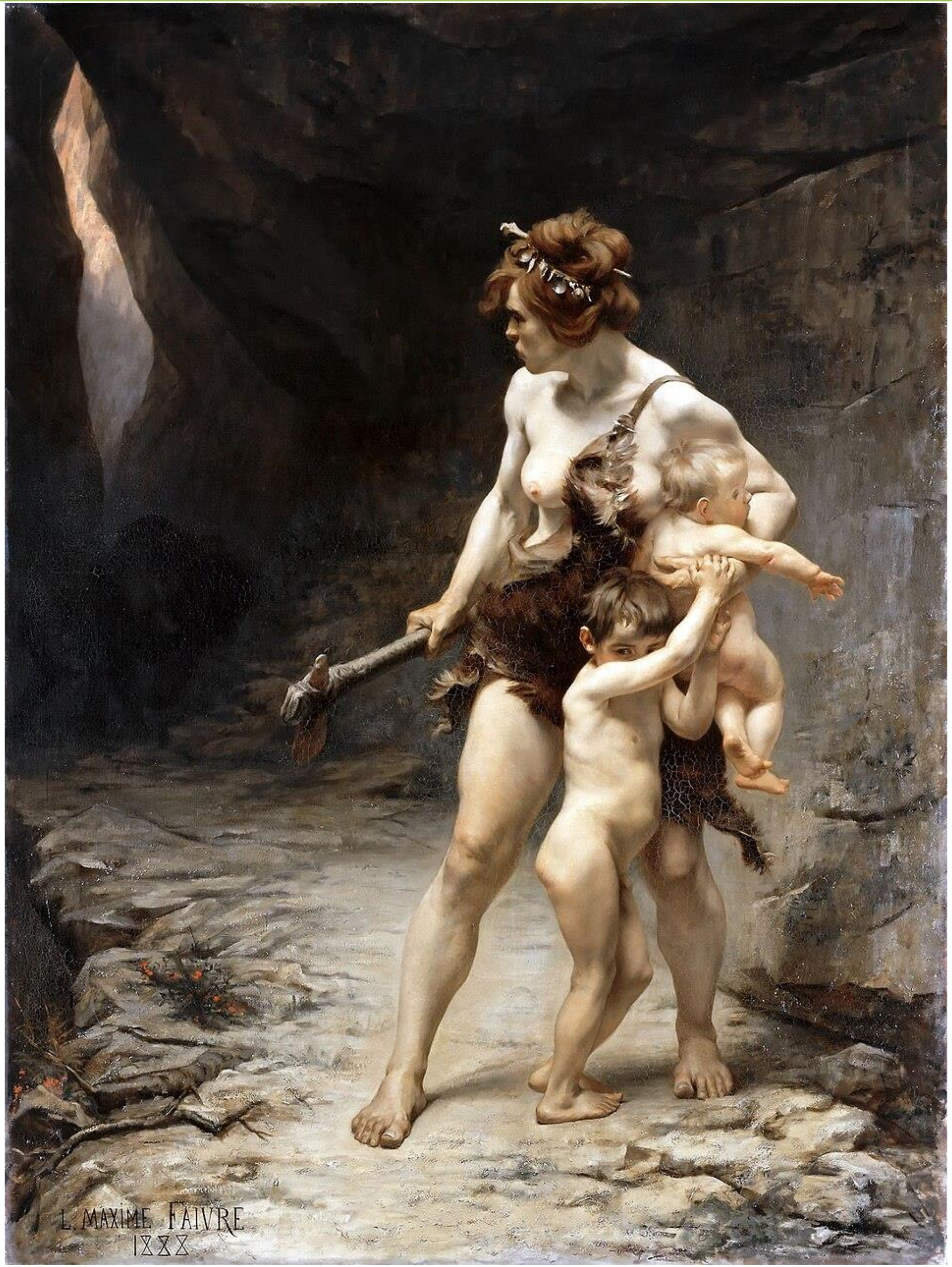


Léon-Maxime Faivre, *Deux mères*, 1988

Life history evolution

Dr. Laura van Holstein

lavanho@emory.edu

Room 215

Part 1:

- Introduction
- Human biology at the species level

Part 2:

- Patterns and drivers of genotypic and phenotypic variation

Part 3:

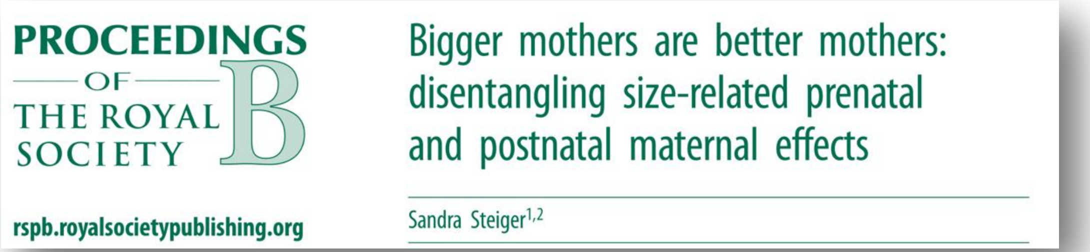
- Individual life course

Outline of today's lecture

- What?
  - What is life history?
  - What is unique about human life history?
- Why?
  - Why does the human life history pattern evolve?
- When?
  - When does the human life history pattern evolve?

Life is about turning energy to offspring

Want to maximise energy input into **all three!**


PROCEEDINGS OF THE ROYAL SOCIETY B

rspb.royalsocietypublishing.org

Bigger mothers are better mothers: disentangling size-related prenatal and postnatal maternal effects

Sandra Steiger[1,2]

Life is about turning energy to offspring

Energy as an evolutionary need needs to be used for reproduction

→ **Growth** — Development

→ **Maintenance** — Group of functions

↳ Maintenance takes a lot of energy

→ **Reproduction**

What?

Why?

When?

Want to maximise energy input into **all three!**

PROCEEDINGS OF THE ROYAL SOCIETY B
rspb.royalsocietypublishing.org

Bigger mothers are better mothers: disentangling size-related prenatal and postnatal maternal effects

Sandra Steiger[1,2]

Life is about turning energy to offspring

But energy is finite...

Growth +

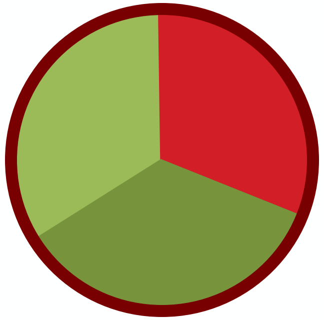
Maintenance +

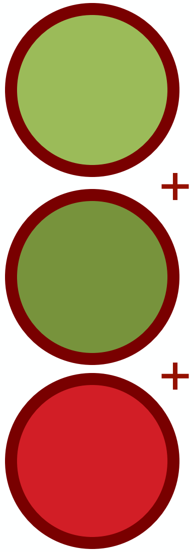
Reproduction

What?
Why?
When?

Want to maximise energy input into **all three!**

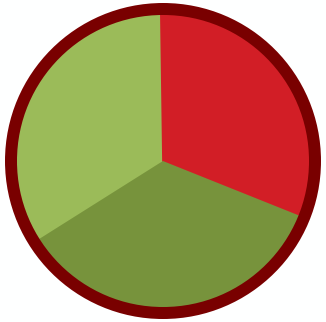
But instead, the optimal balance must be found:





No Darwinian demons!

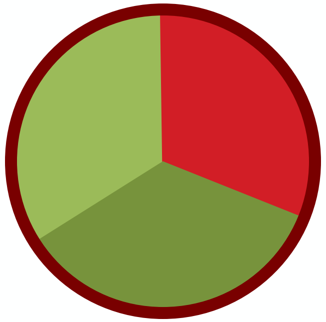
But instead, the optimal balance must be found:



Crux of life history theory:

tradeoffs

Why not the same balance in every species?



What?
Why?
When?

Why not the same balance in every species? The environment!

→ Animals don't live in the same environments

Unstable environments; high mortality

→ Under Charnov's model, earlier alpha

Growth | Reproduction

- Many infants in litter
- Precocial infants — can do a lot already
- Short growth period
- Little parental care
- Short interbirth interval (IBI)
- Many babies over lifetime

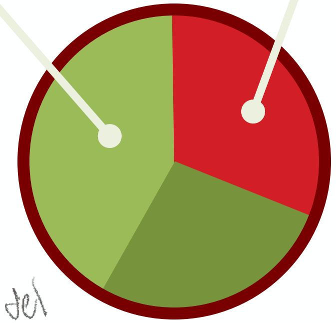
r → population growth as much as possible

Stable environments; low mortality

→ later alpha

Growth | Reproduction

low mortality allows for greater investment into growth under Charnov's model →

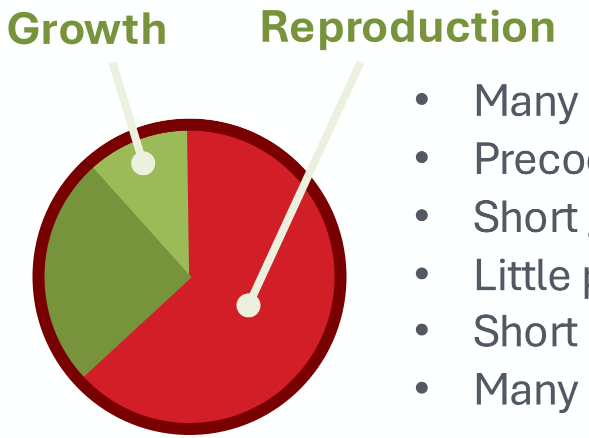


- Few infants in litter
- Altricial infants
- Long growth period
- Lots of parental care
- Long IBI
- Few babies over lifetime

K → population size sticks to carrying capacity

Why not the same balance in every species? The environment!

Unstable environments; high mortality



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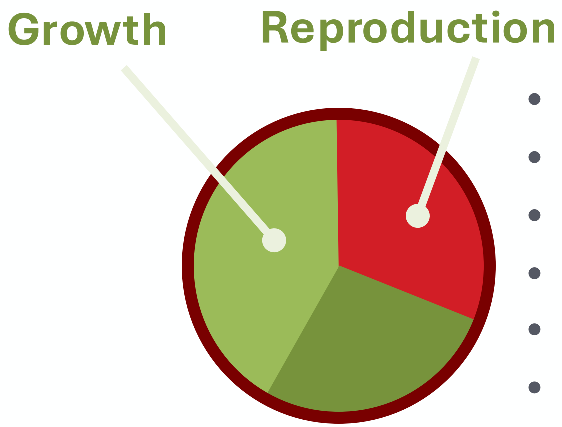

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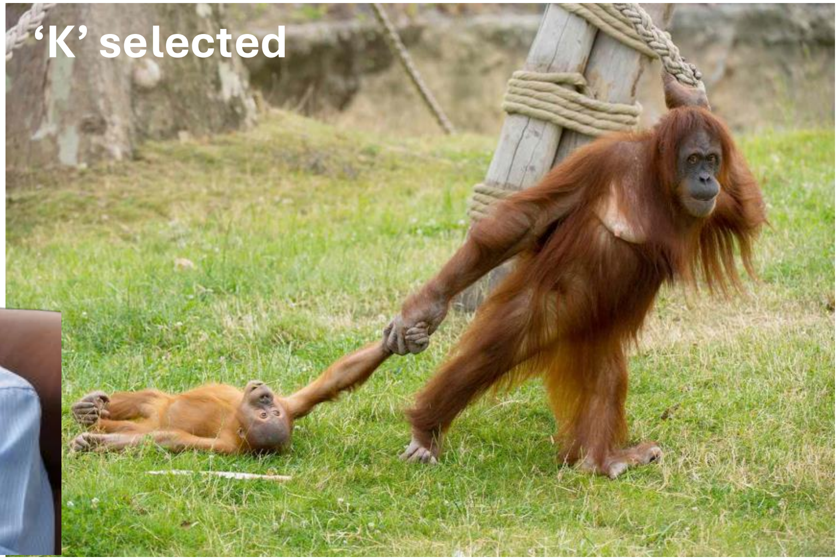

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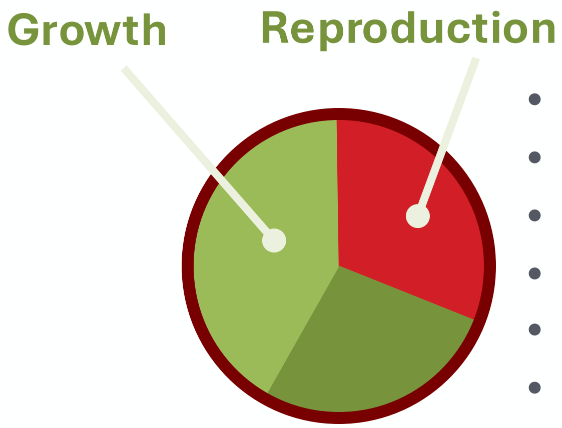

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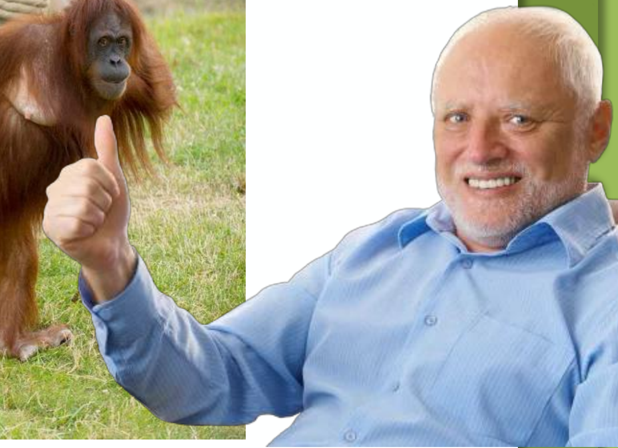

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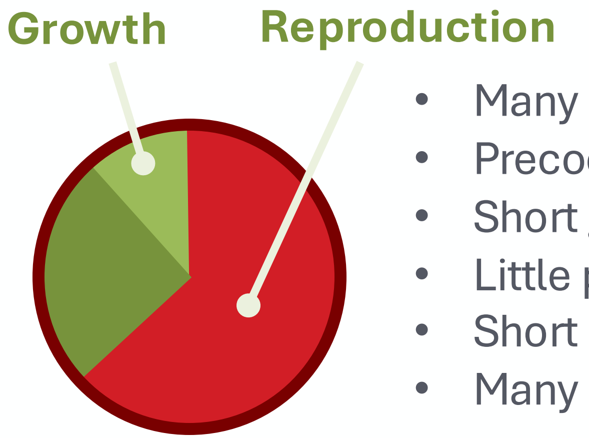

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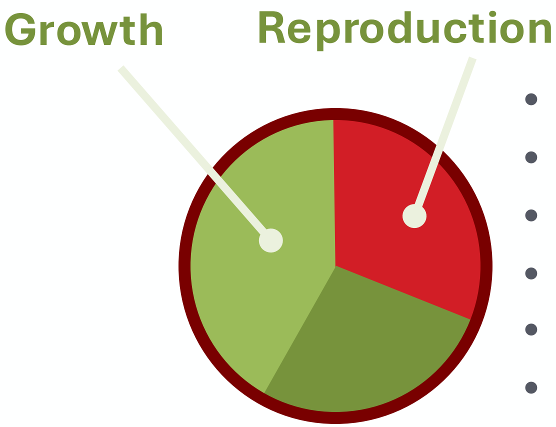

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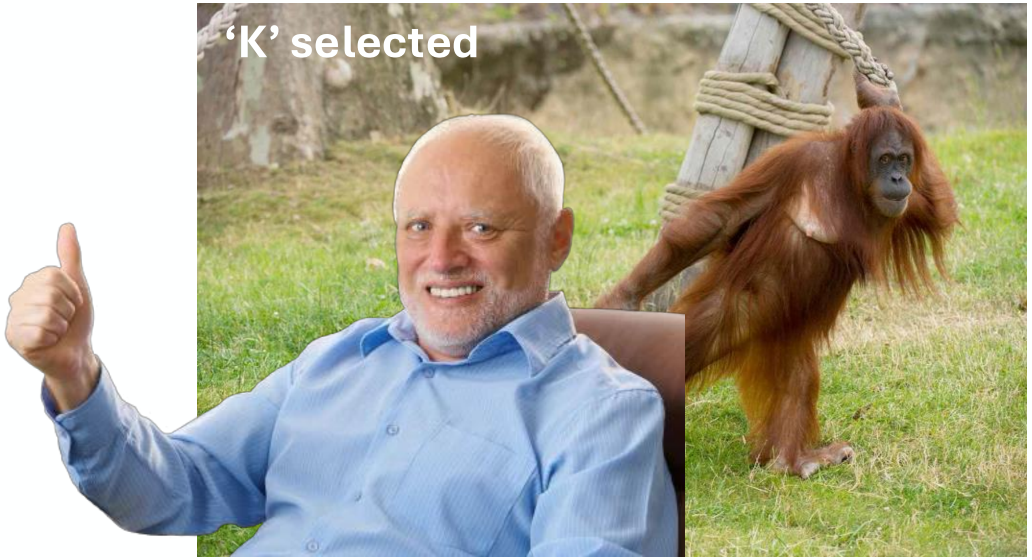
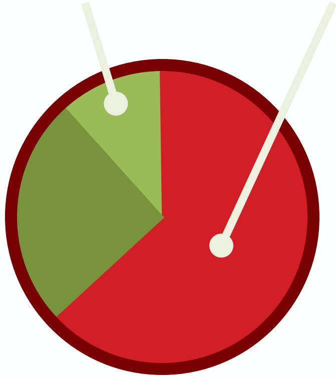


Why not the same balance in every species? The environment!

what about humans?

Unstable environments; high mortality

Growth | Reproduction

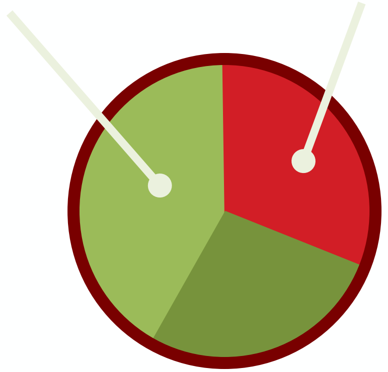


- Many infants in litter
- Precocial infants
- Short growth period
- Little parental care
- **Short interbirth interval (IBI)**
- **Many babies over lifetime**

High fertility care

Stable environments; low mortality

Growth | Reproduction



- **Few infants in litter**
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What?
Why?
When?

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Growth Reproduction

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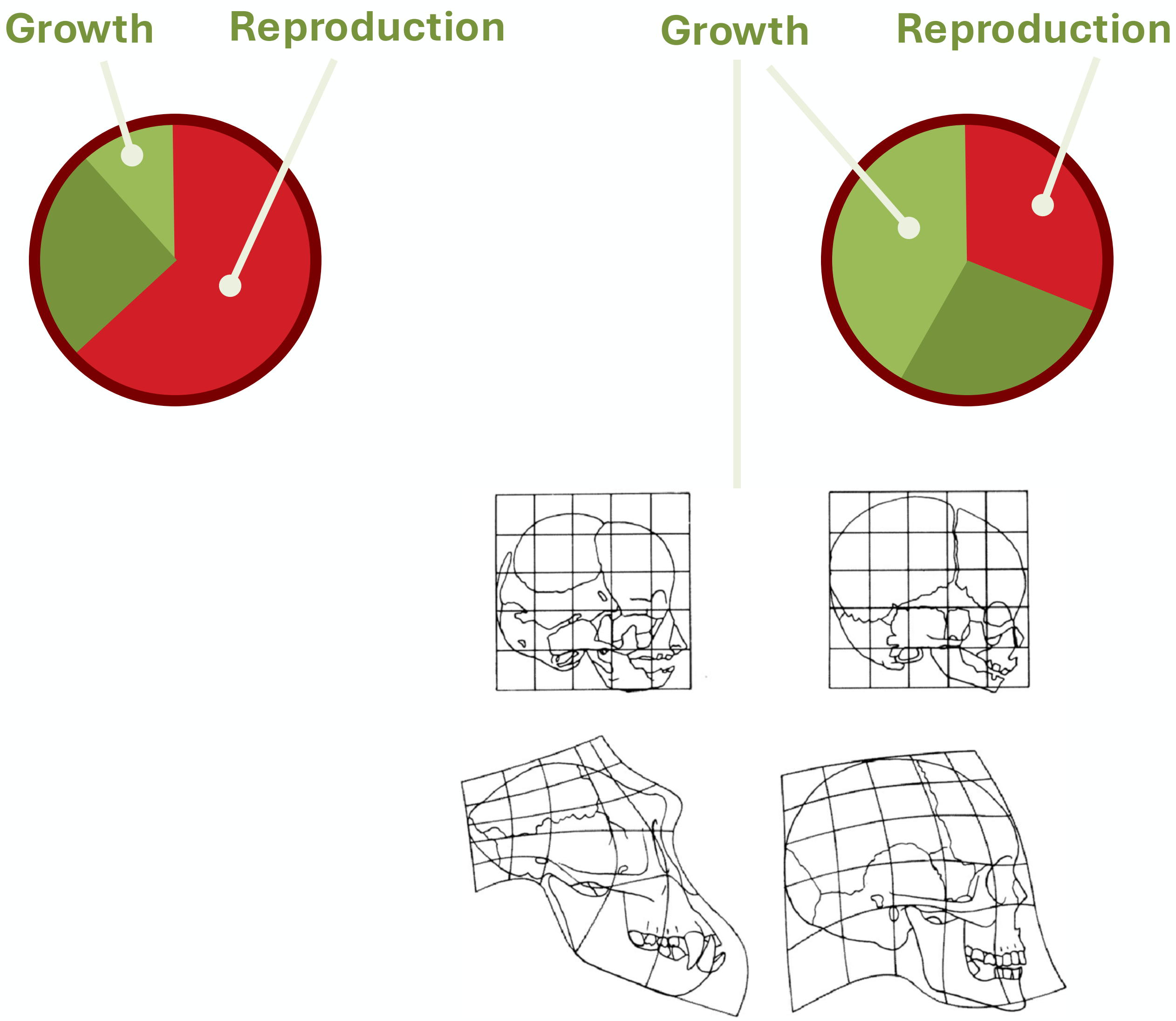
What?
Why?
When?

Tradeoffs in life history theory:

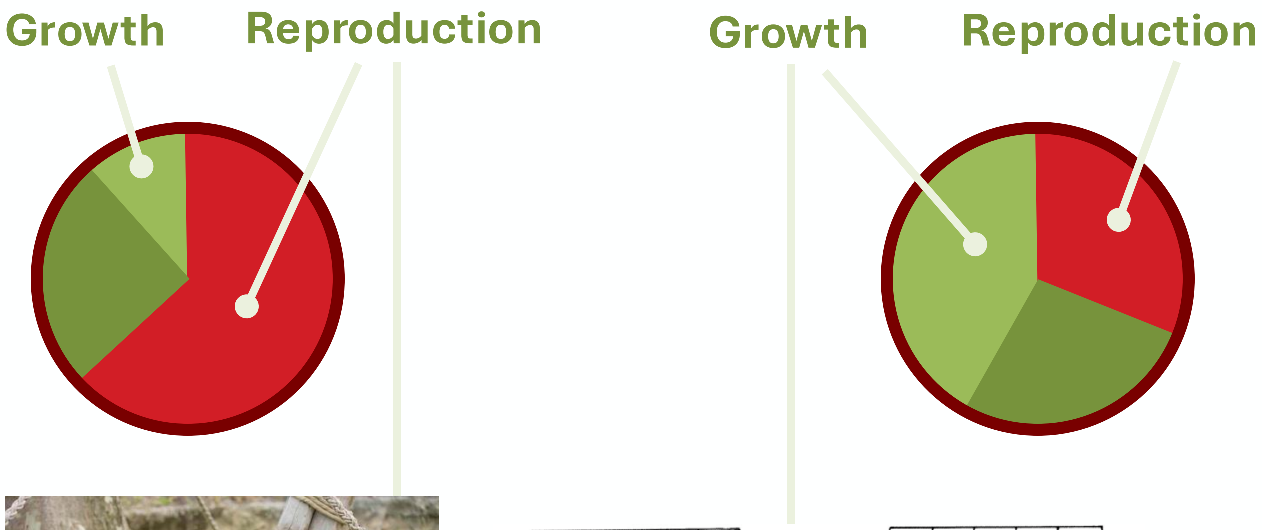


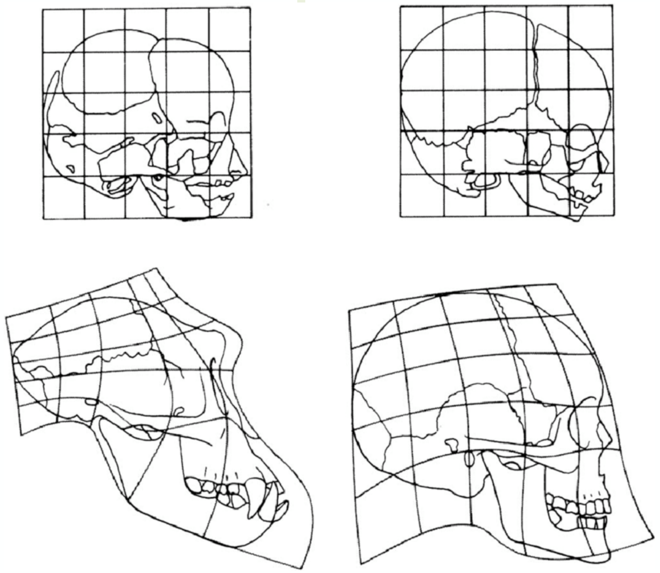
1. **Classic tradeoffs between growth, maintenance, and reproduction**
   (r-K continuum)

2. **Growth of different body parts**

What?
Why?
When?

Tradeoffs in life history theory:

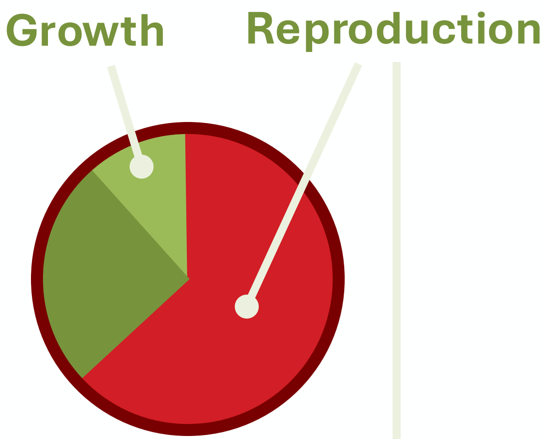


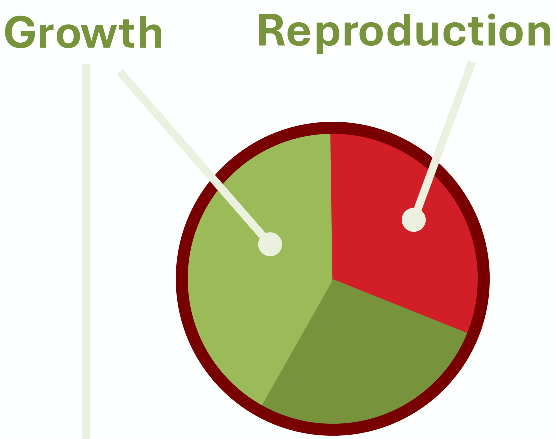


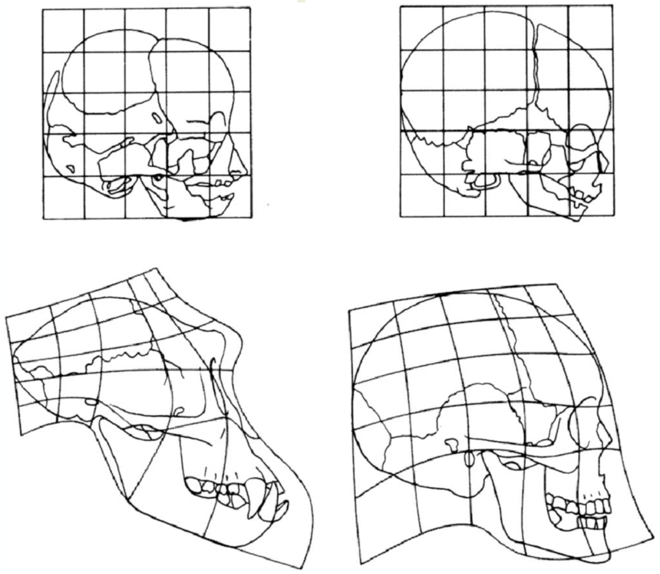
1. **Classic tradeoffs between growth, maintenance, and reproduction**
   (r-K continuum)

2. **Growth of different body parts** → growth tradeoffs
3. **Quantity-quality tradeoff**
   ↳ Just reproduction - many babies or 1 high quality baby?

Tradeoffs in life history theory:







1. **Classic tradeoffs between growth, maintenance, and reproduction**
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4. **Investment in current vs future offspring**
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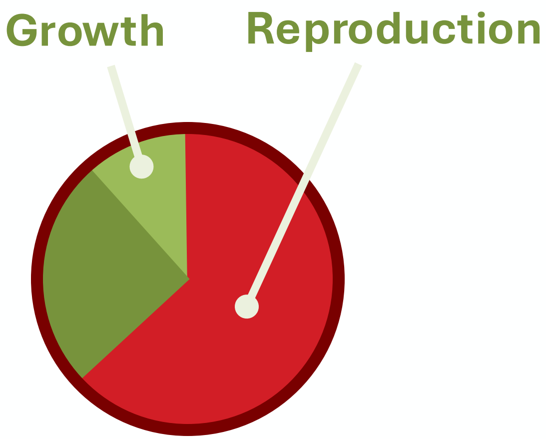
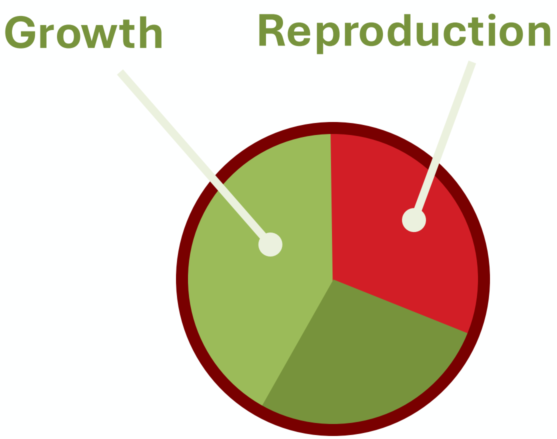
6. **Growth rates**
7. **Immune response vs normal metabolism**
8. **Energy use of different organs**

etc!

Tradeoffs in life history theory:





1. **Classic tradeoffs between growth, maintenance, and reproduction** (r-K continuum)

Proportion of total daily energy allocated to growth:
- 28-39% in first 3 months of infancy (so 61-72% to maintenance)
- <2% at age 2-6 (so 98% to maintenance)

40% of total daily energy to the brain at age 5

2. **Growth of different body parts**
3. **Quantity-quality tradeoff**
4. **Investment in current vs future offspring**
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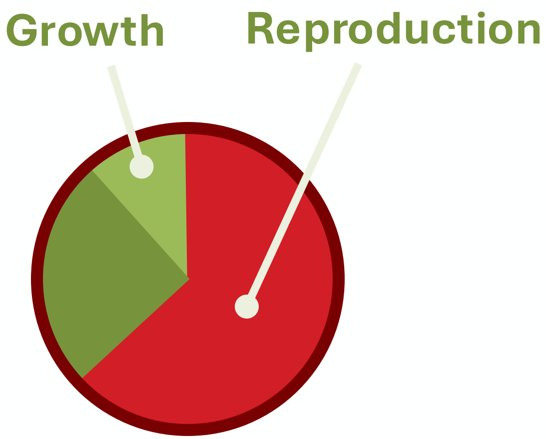

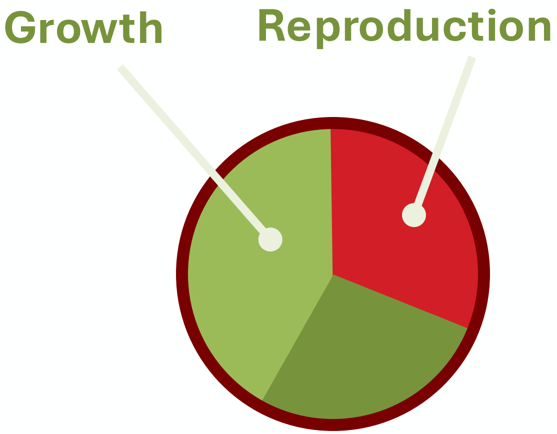

Key

Allocations vary across the lifespan

Distinct stages can be identified: these differ from each other in their relative allocations

For example:

Infancy: growth + maintenance, no reproduction

1. **Classic tradeoffs between growth, maintenance, and reproduction**
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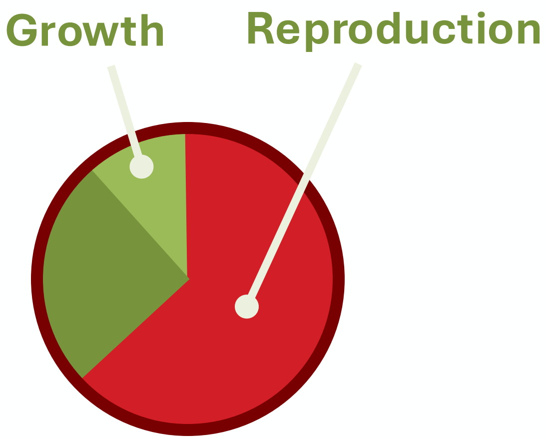

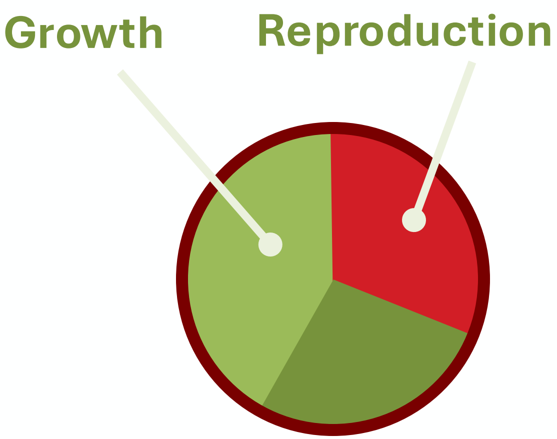

Key

Allocations vary across the lifespan

Distinct stages can be identified: these differ from each other in their relative allocations

For example:

Adulthood: maintenance + reproduction, no growth

1. **Classic tradeoffs between growth, maintenance, and reproduction** (r-K continuum)
2. **Growth of different body parts**
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4. **Investment in current vs future offspring**
5. **Investment in each offspring**
6. **Growth rates**
7. **Immune response vs normal metabolism**
8. **Energy use of different organs**

etc!

What?
Why?
When?

A generalized mammalian life history

Uterine growth

Infancy

point of transition btwn infancy and juvenile period

Juvenile

No childhood for non humans - juveniles

Adulthood

Transition btwn juvenile period & adulthood

Birth

Marks transition btwn uterine growth & infancy

First molar erupts

M1 eruption
Weaning
Forage independently

First reproduction

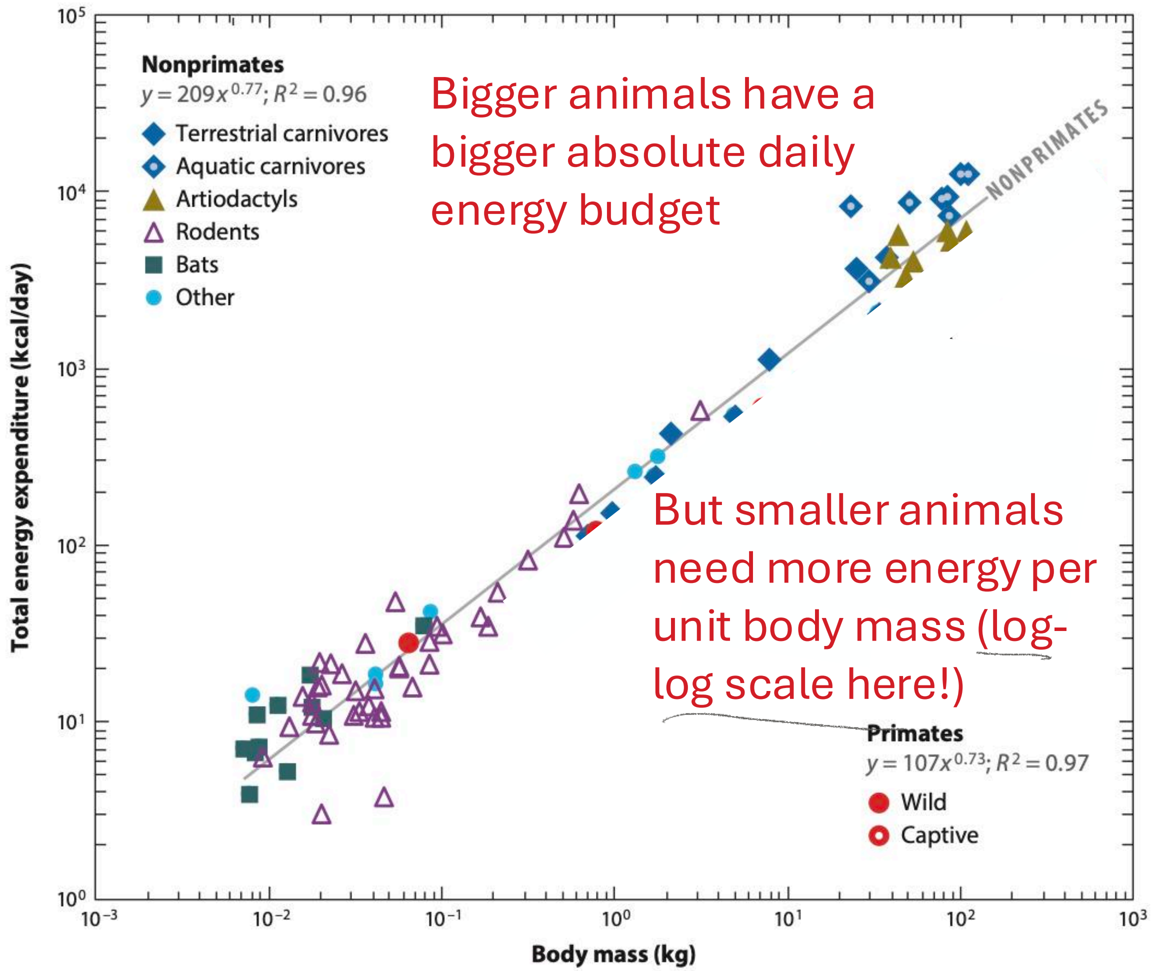
High rate of growth

Slow rate of growth

learning

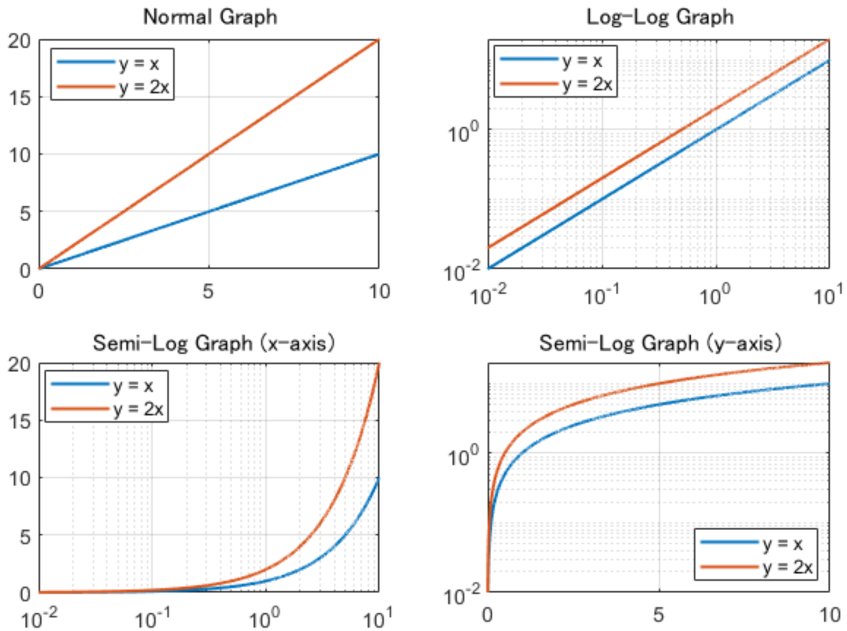
Maintenance and reproduction

A generalized mammalian life history



Bigger animals have a bigger absolute daily energy budget

But smaller animals need more energy per unit body mass (log-log scale here!)



Maintenance and reproduction

Pontzer (2015)

Primates start breaking this generalized mammalian life history

Uterine growth | Infancy | Juvenile | Adulthood

Birth

M1 eruption
Weaning
Forage independently

First reproduction

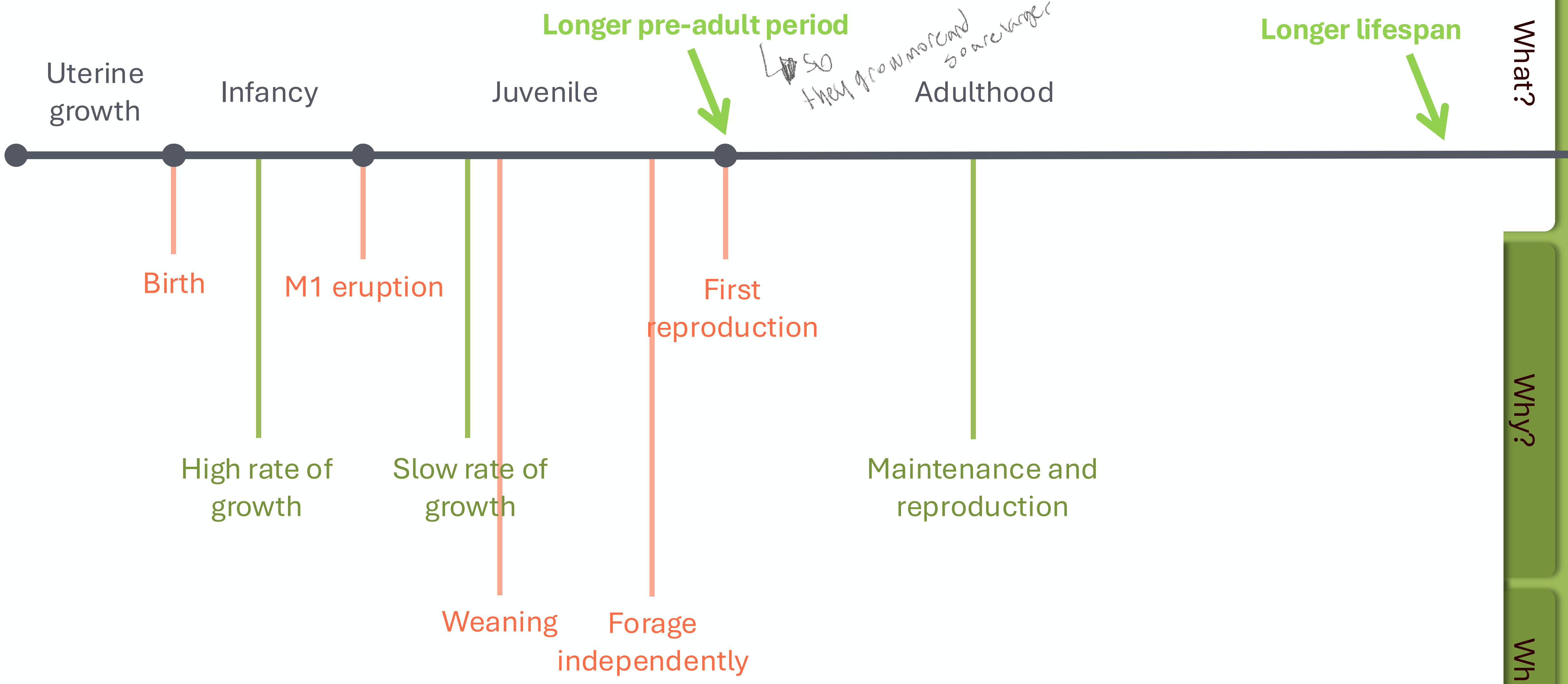
High rate of growth

Slow rate of growth

Maintenance and reproduction

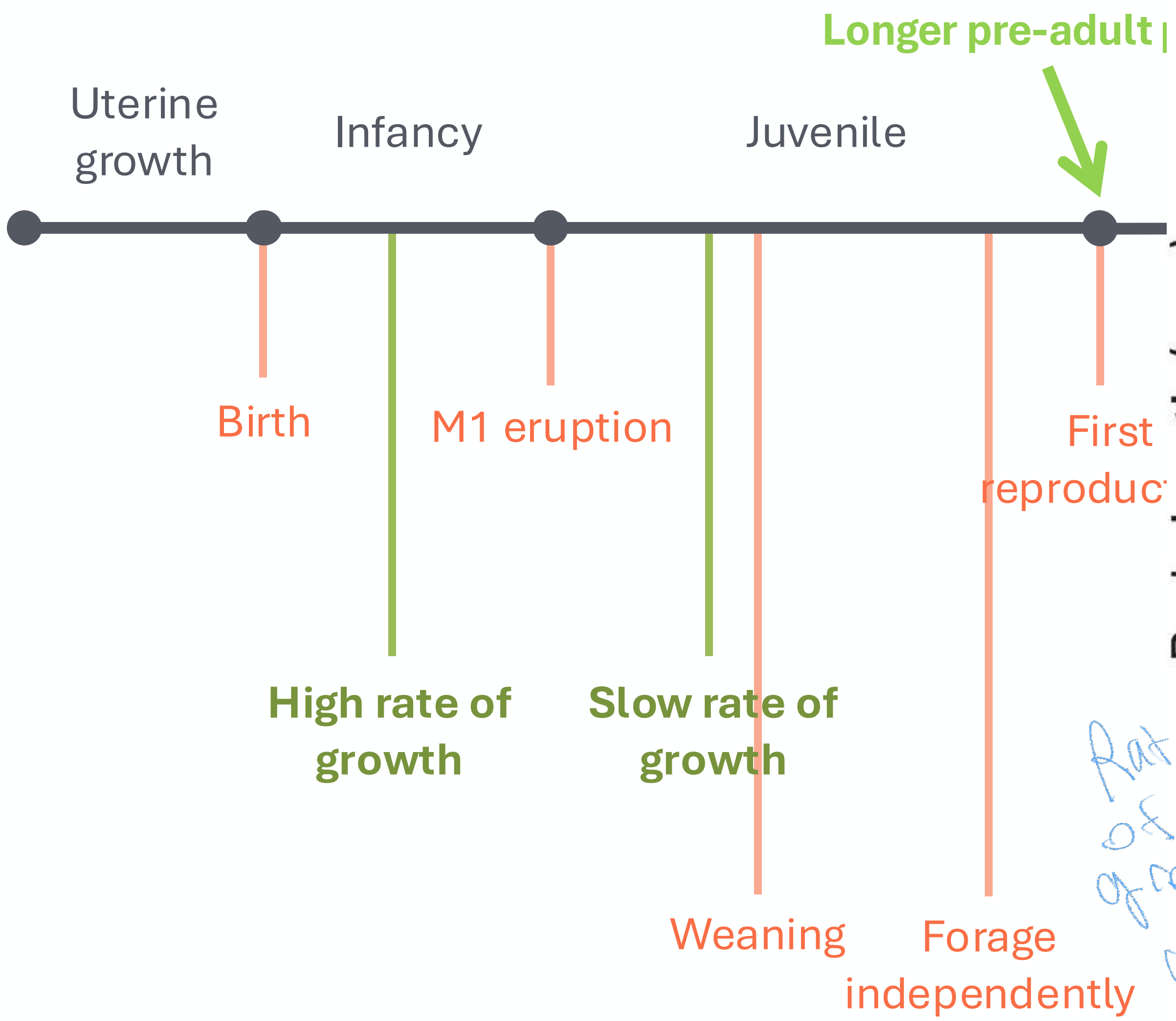
What?
Why?
When?

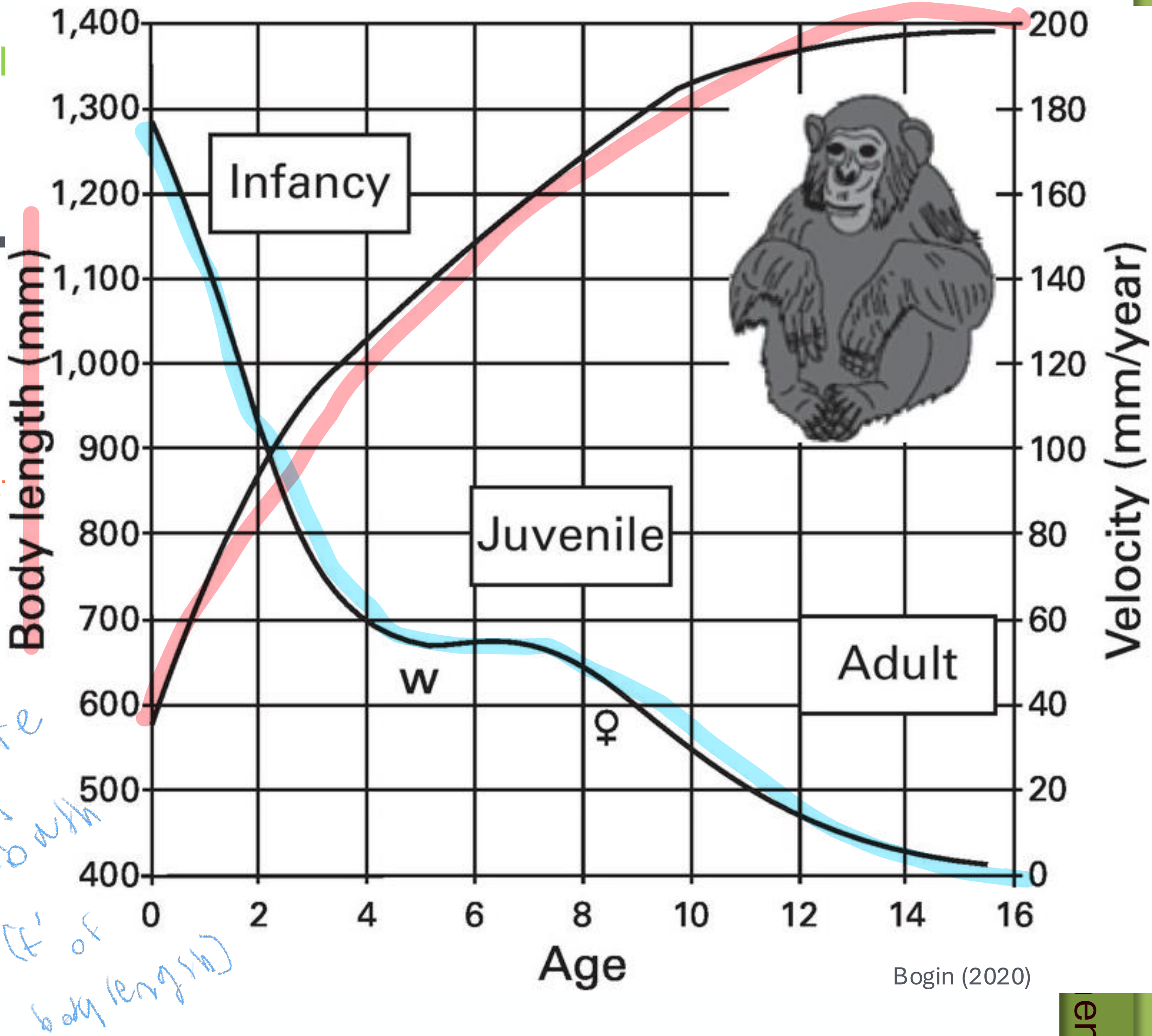
Primates start breaking this generalized mammalian life history



What?
Why?
When?

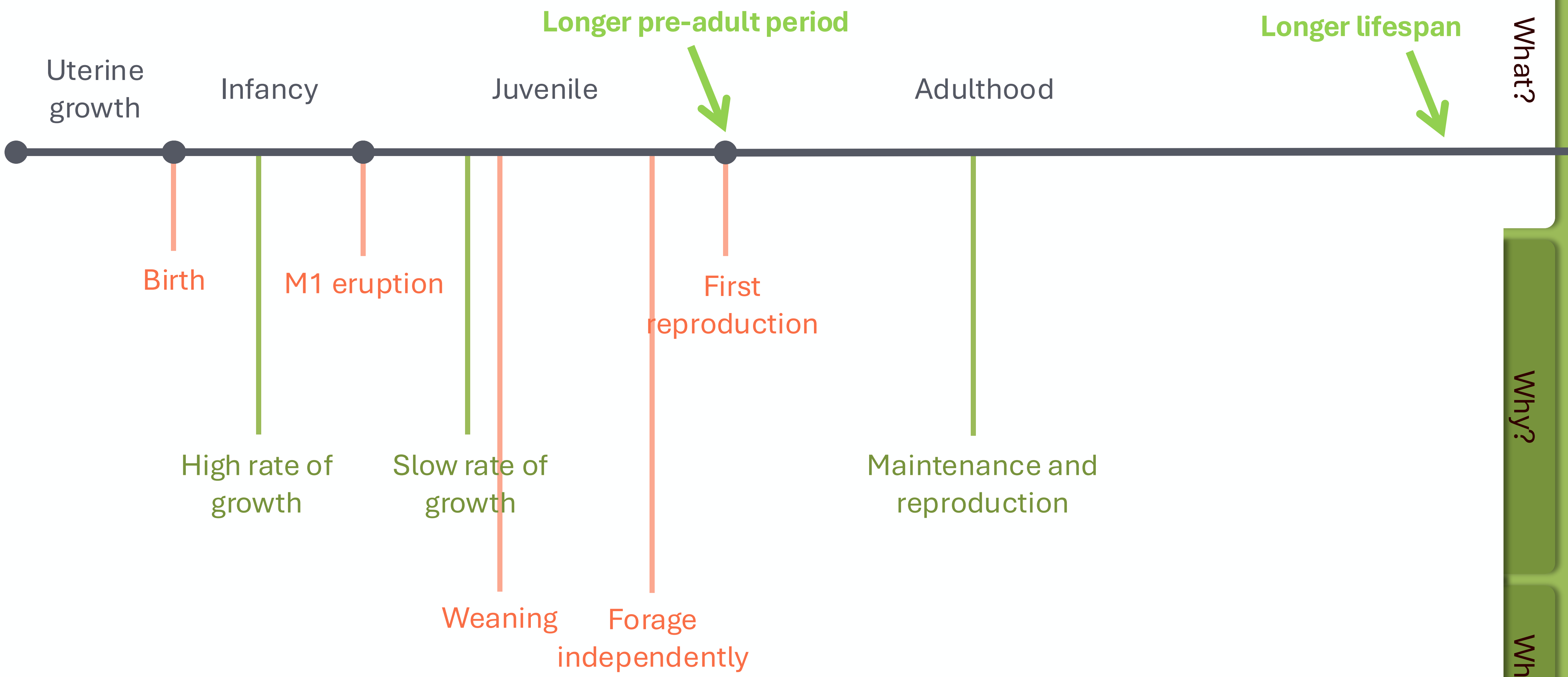
Primates start breaking this generalized mammalian life history



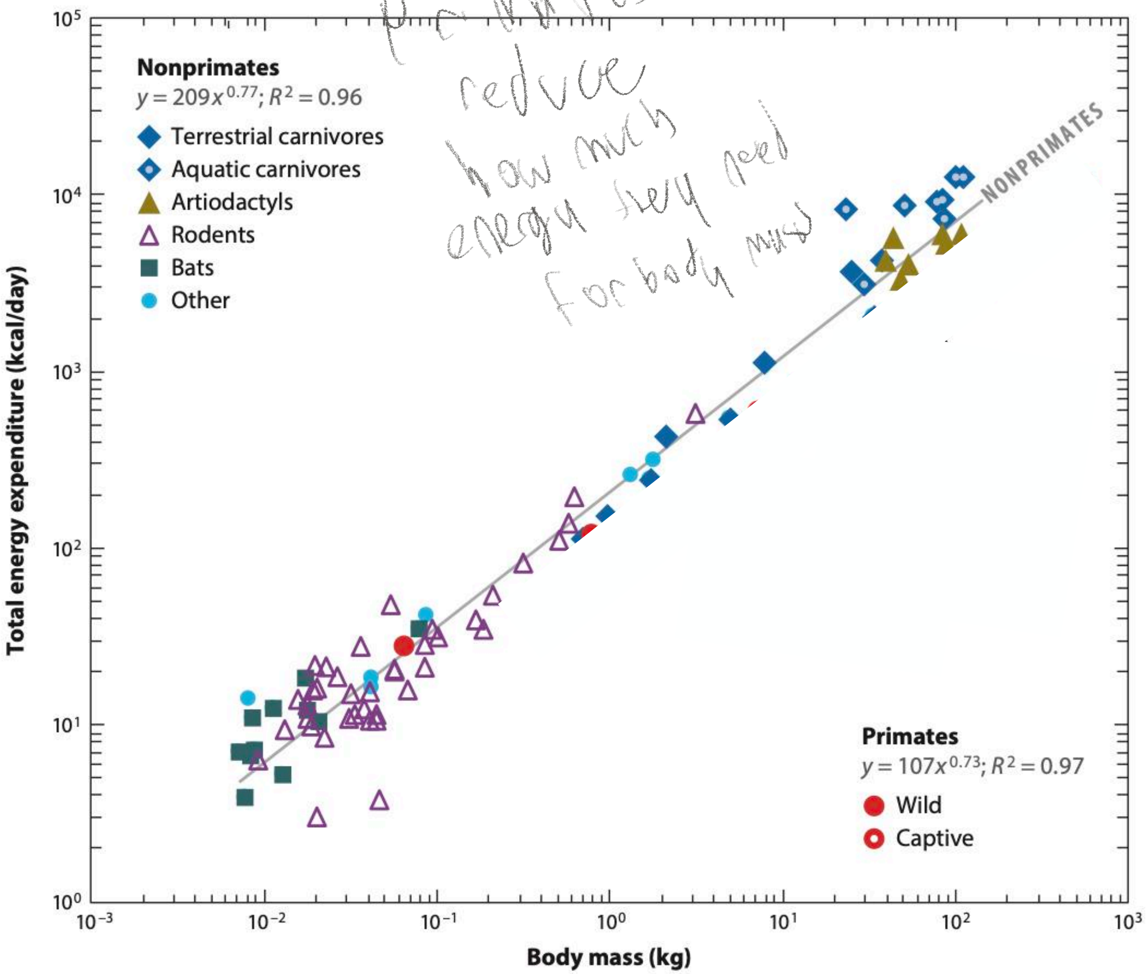


Bogin (2020)

Primates start breaking this generalized mammalian life history



Primates start breaking this generalized mammalian life history



Primates reduce how much energy they need for body mass

Longer pre-adult period

Longer lifespan

Adulthood

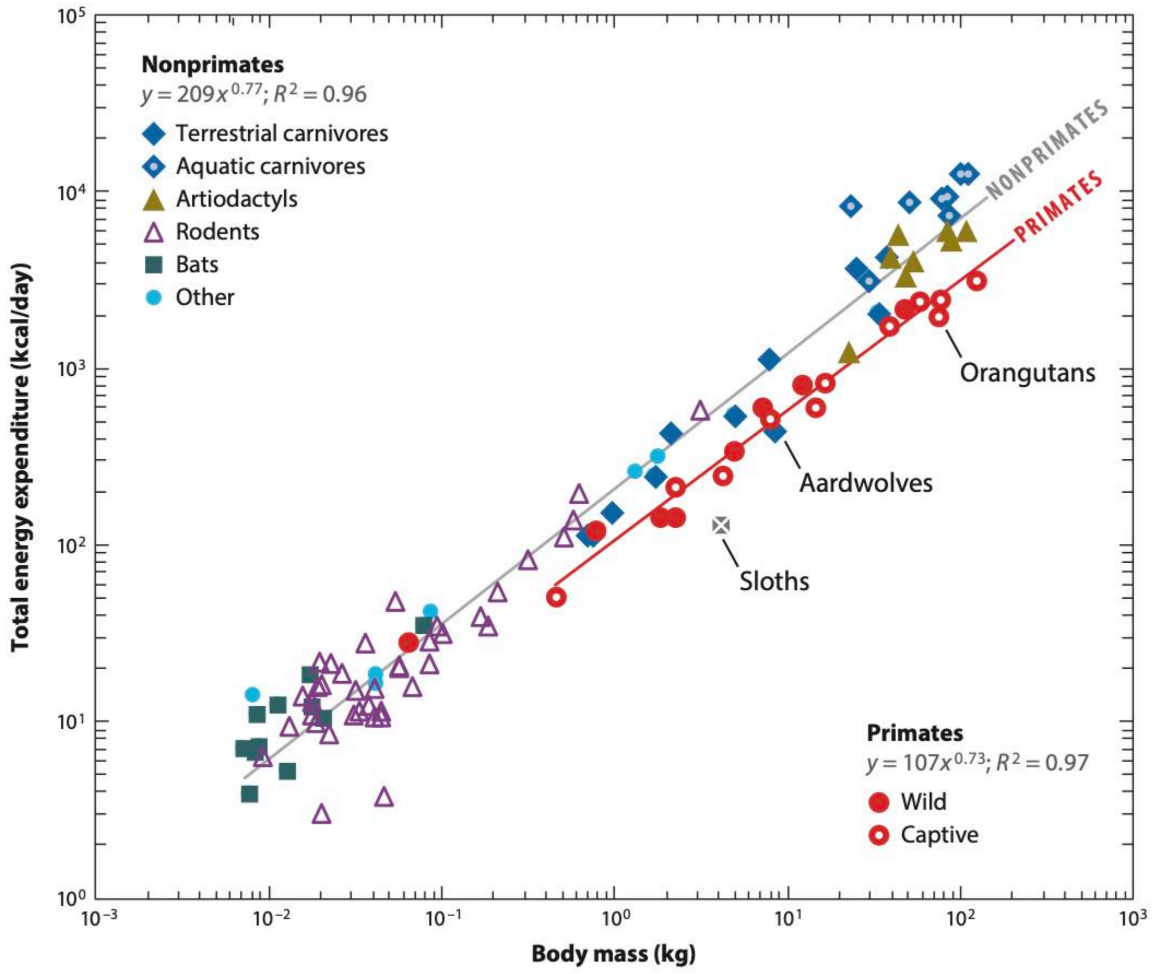
Maintenance and reproduction

Primates retain life history stages

What?
Why?
When?

Pontzer (2015)

Primates start breaking this generalized mammalian life history



Pontzer (2015)

Longer pre-adult period

Longer lifespan

Adulthood

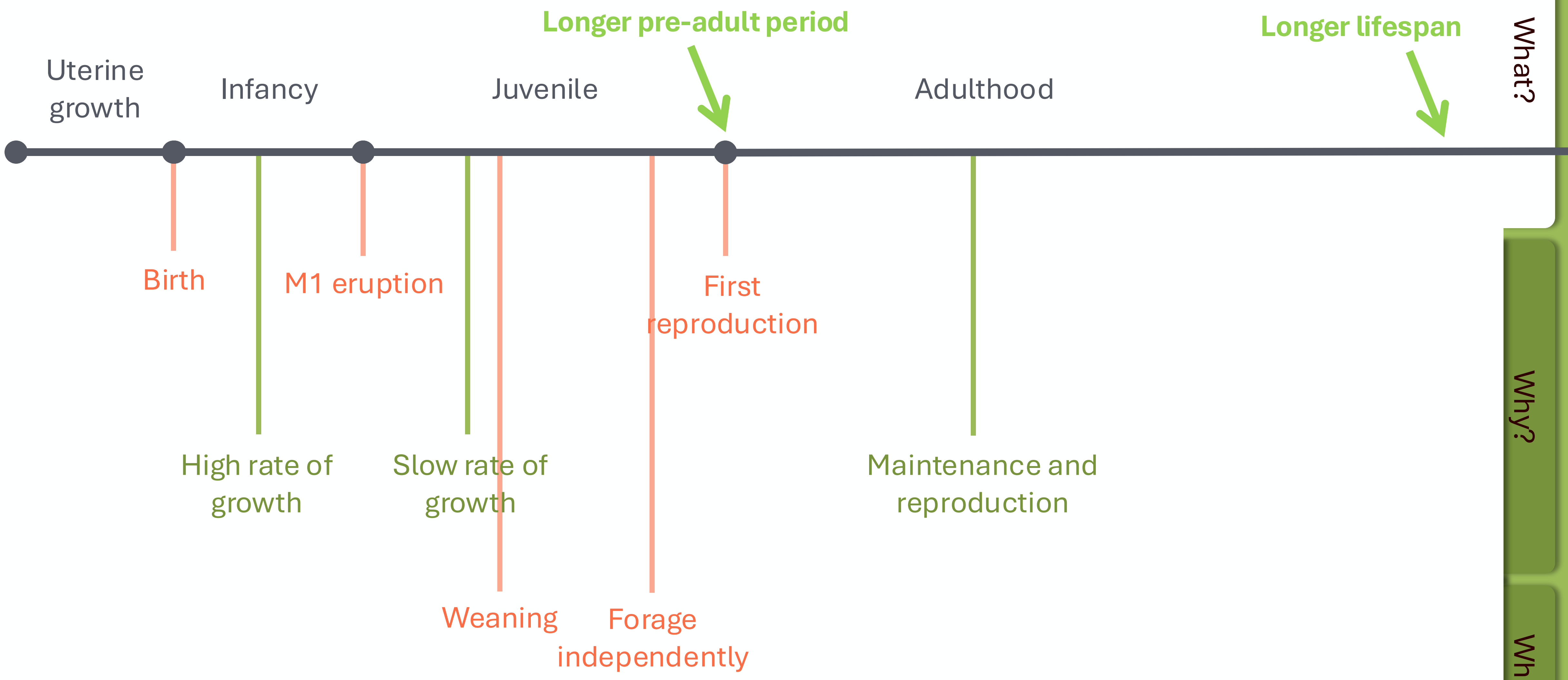
Maintenance and reproduction

Primates use 50% less energy per day than other mammals of their body mass

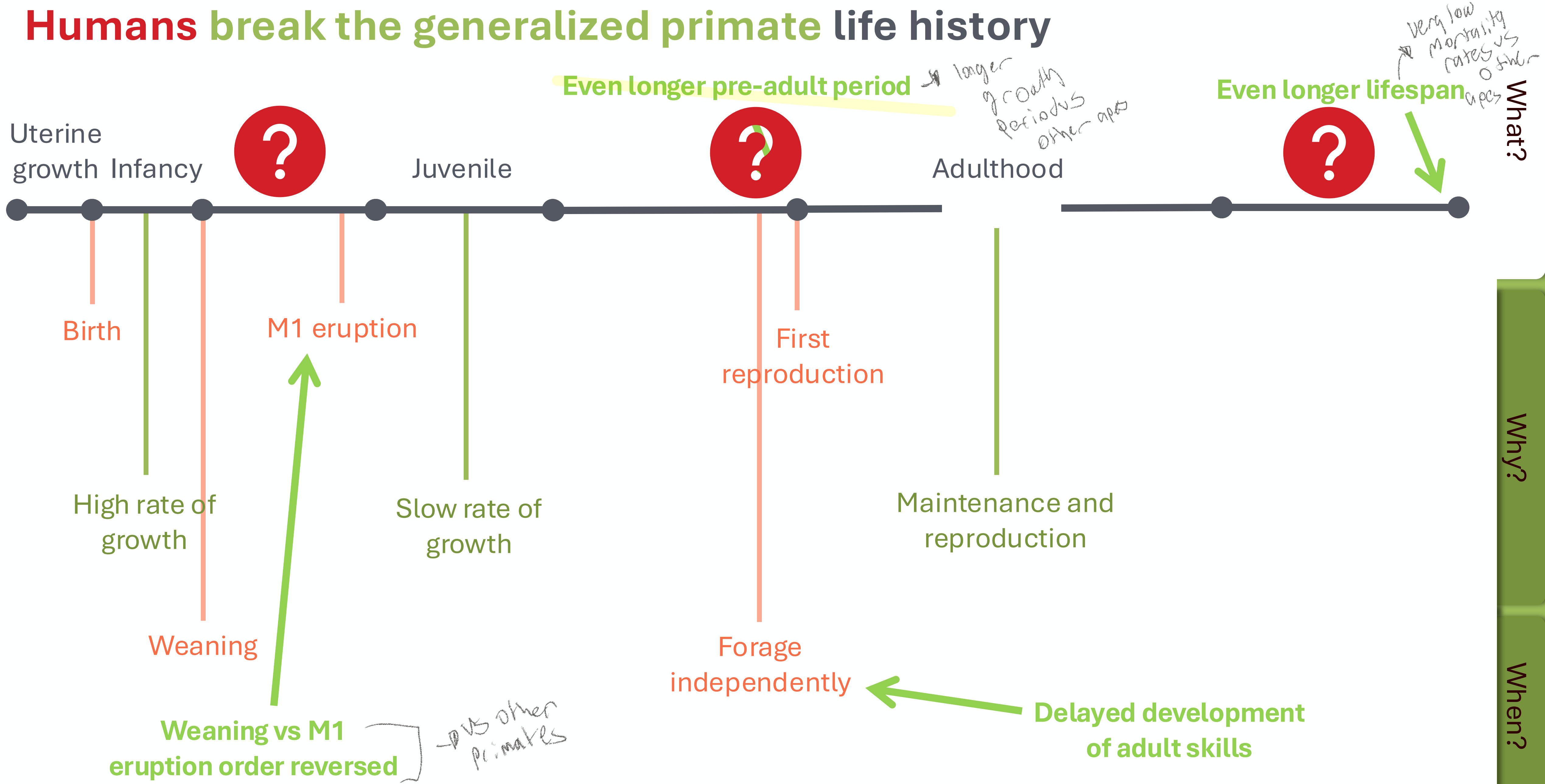
This has been argued by Pontzer to underlie their exceptionally slow life histories!

What? Why? When?

Primates start breaking this generalized mammalian life history



Humans break the generalized primate life history



What?
Why?
When?

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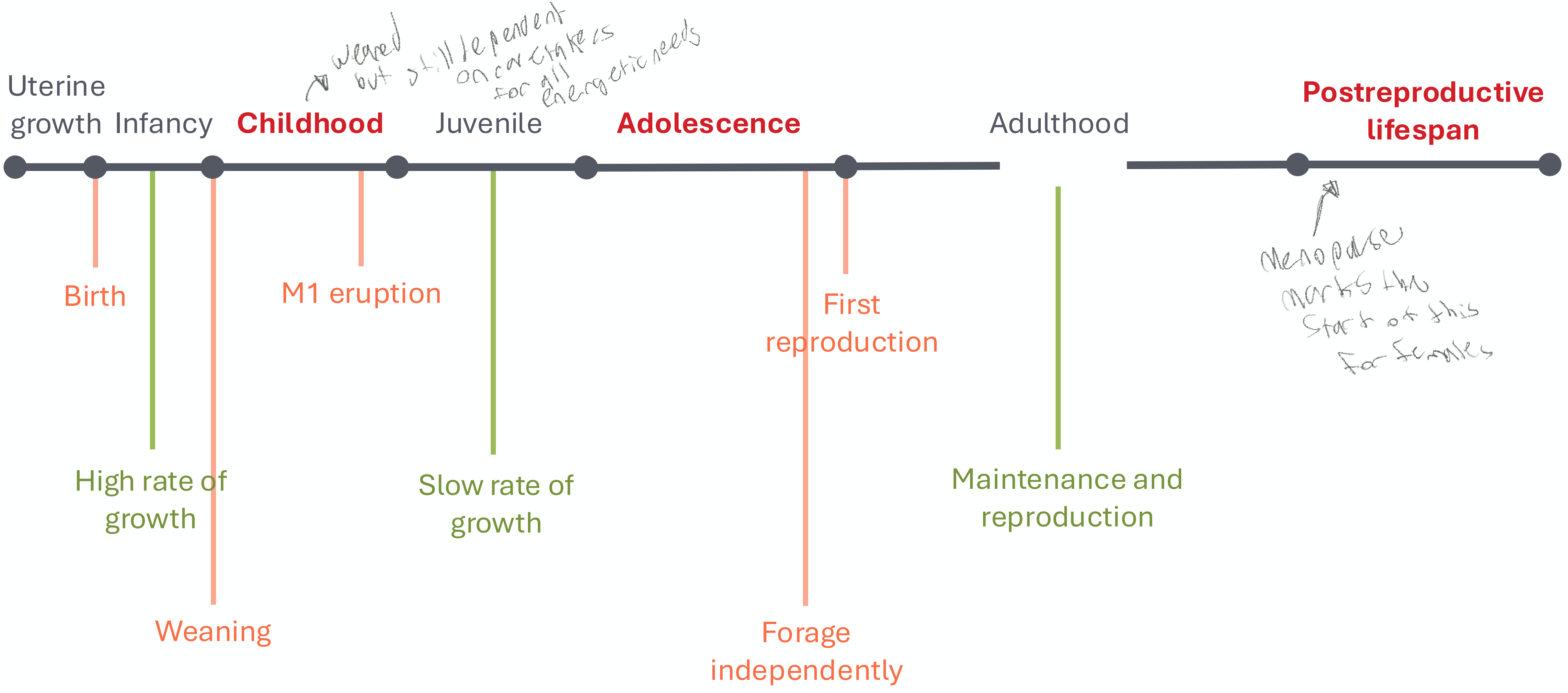
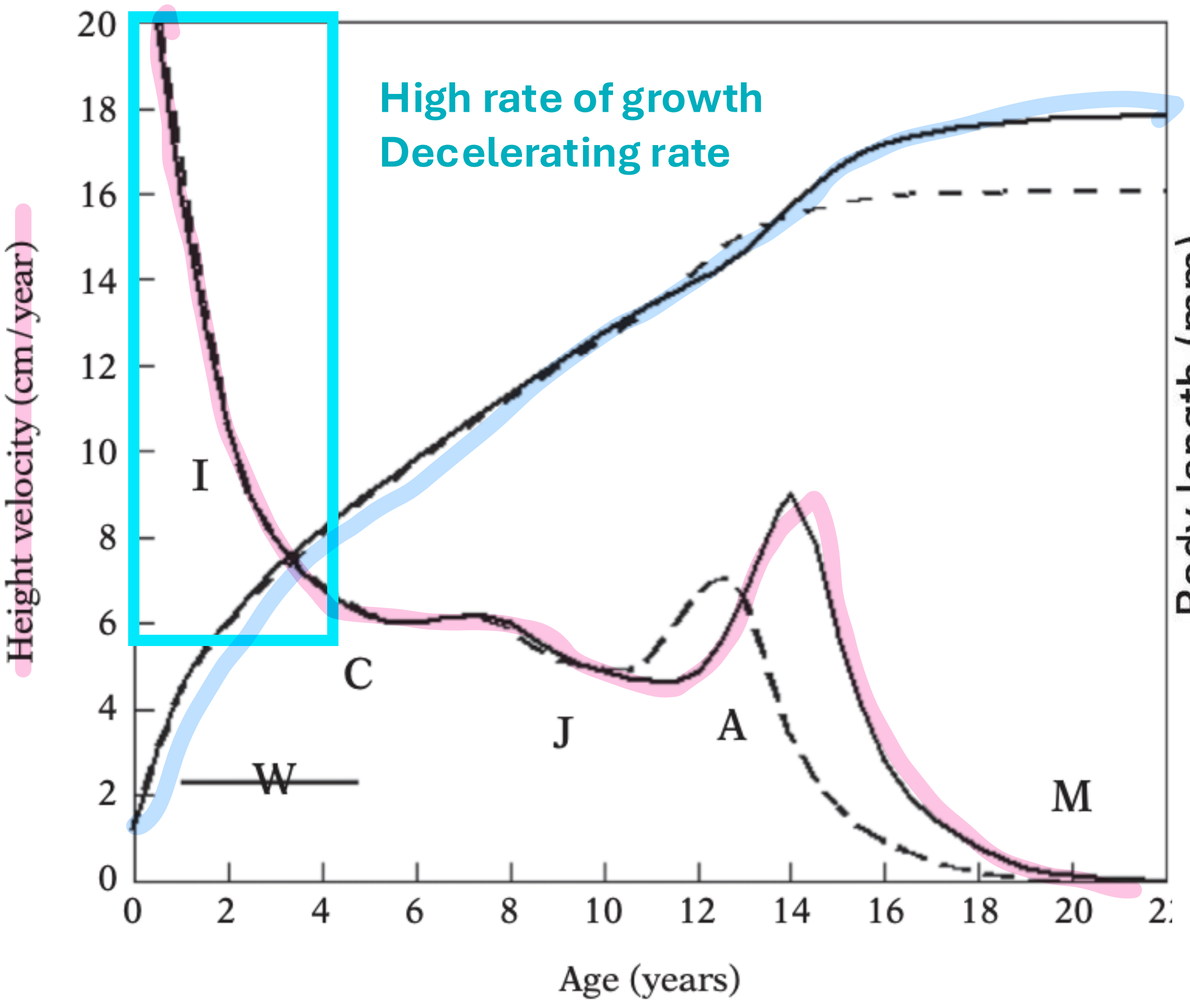
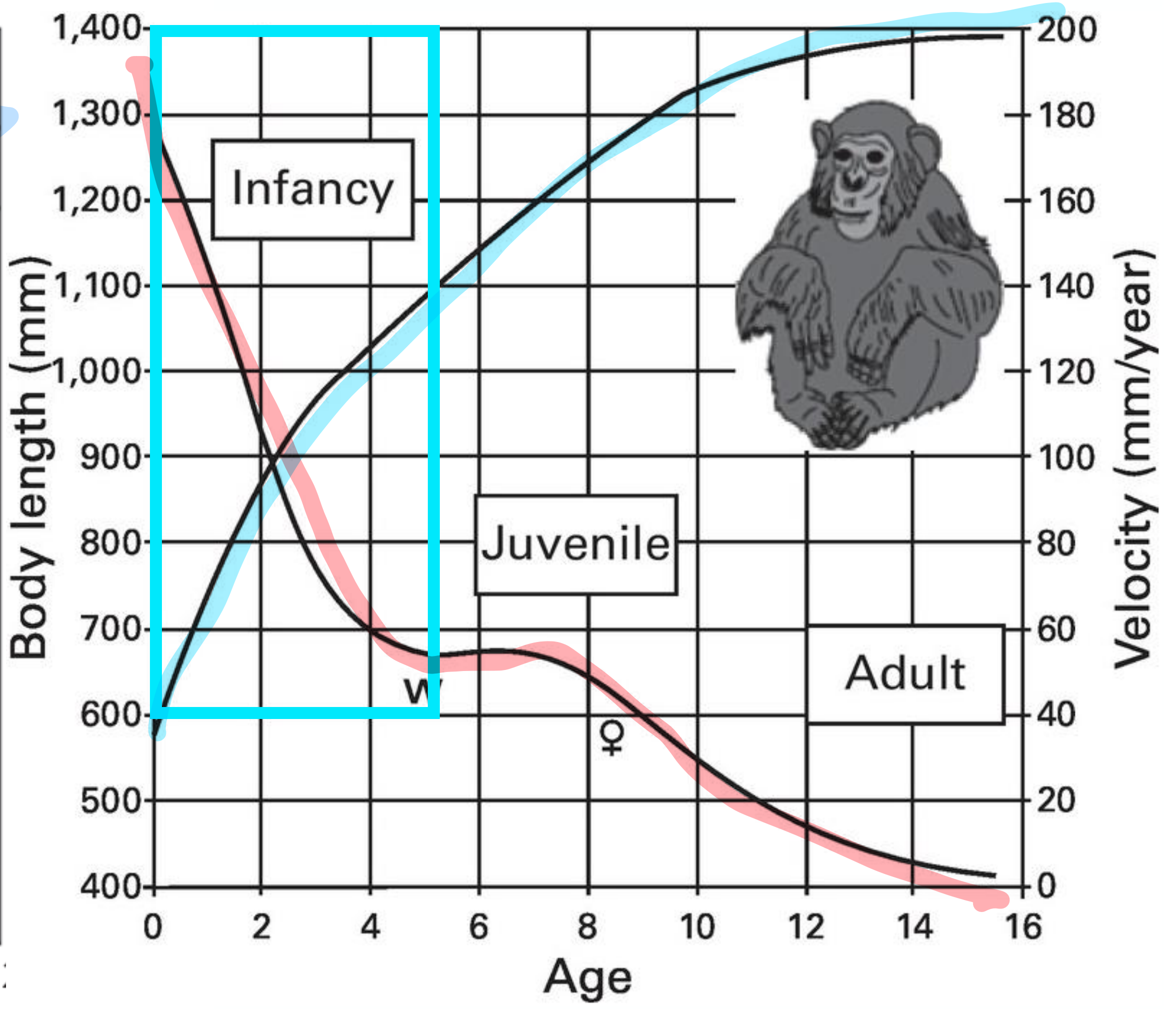


Humans break the generalized primate life history

High rate of growth
Decelerating rate





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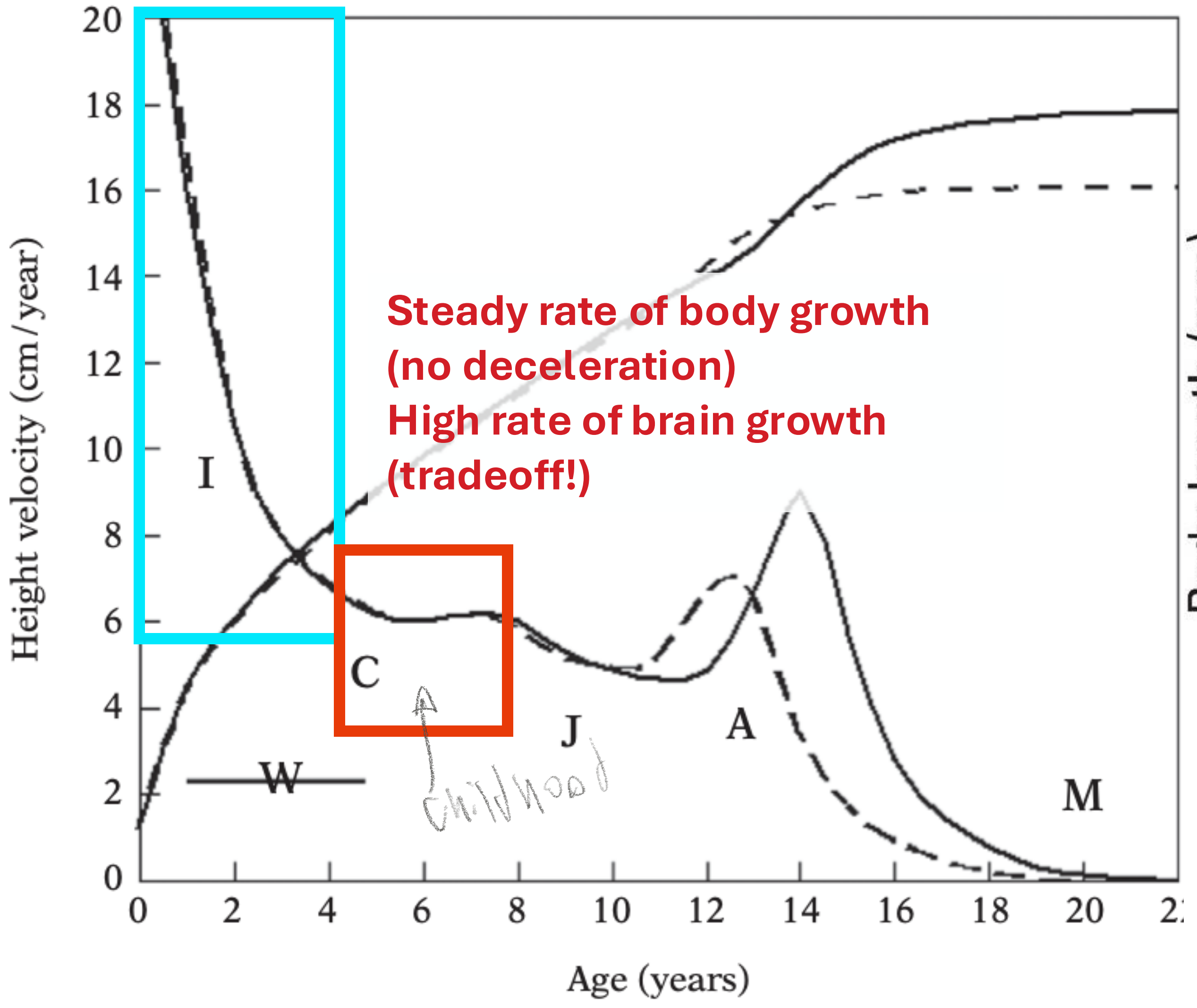

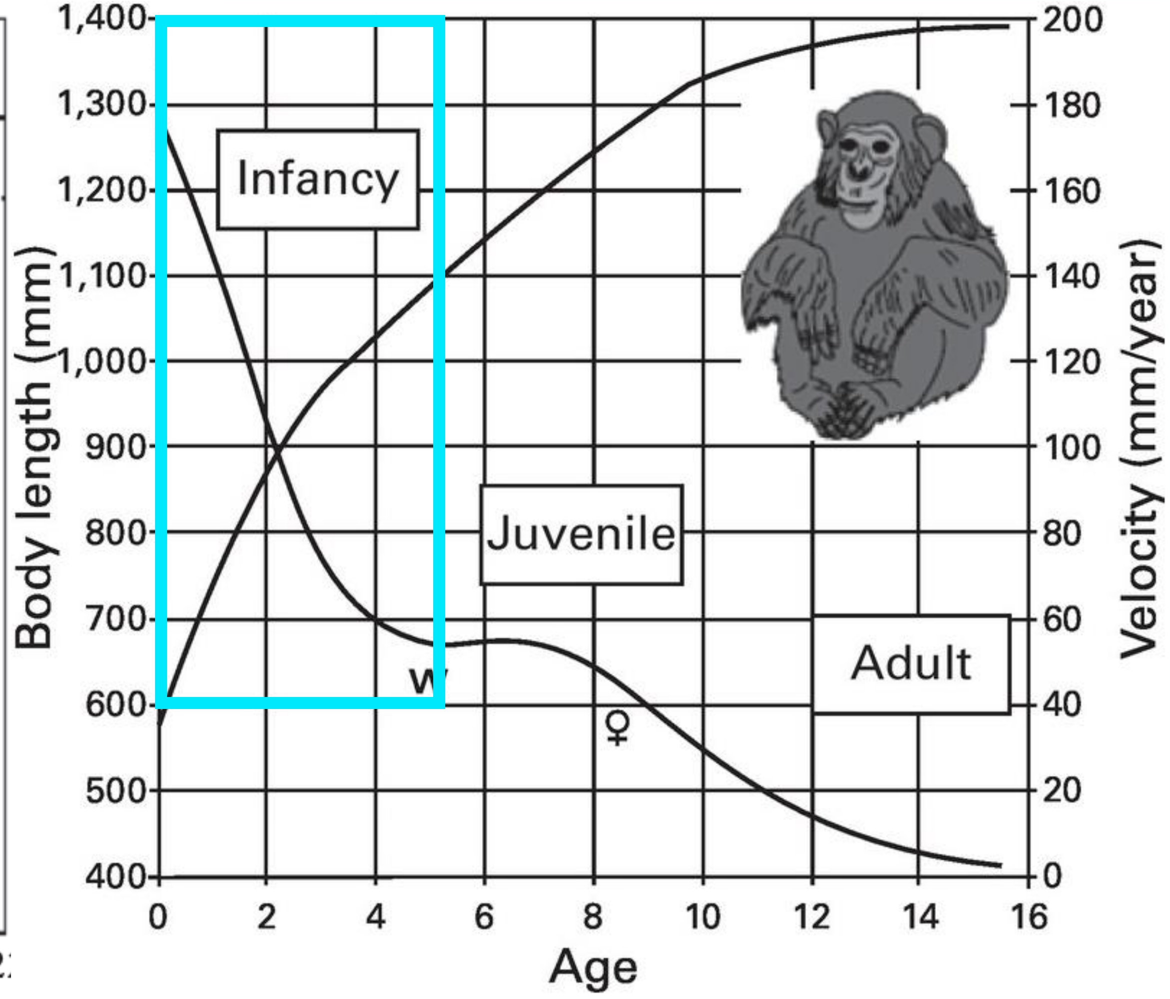

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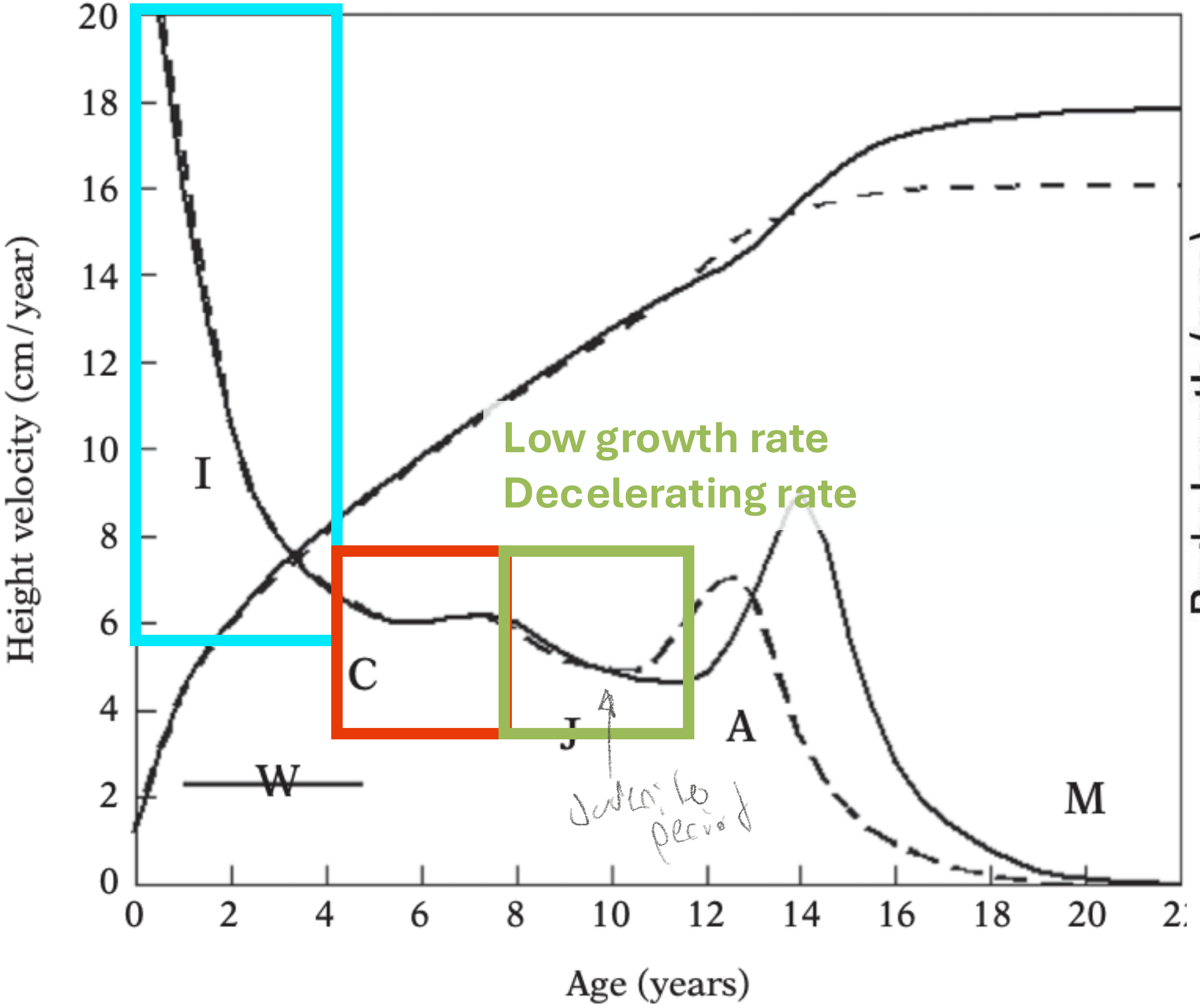

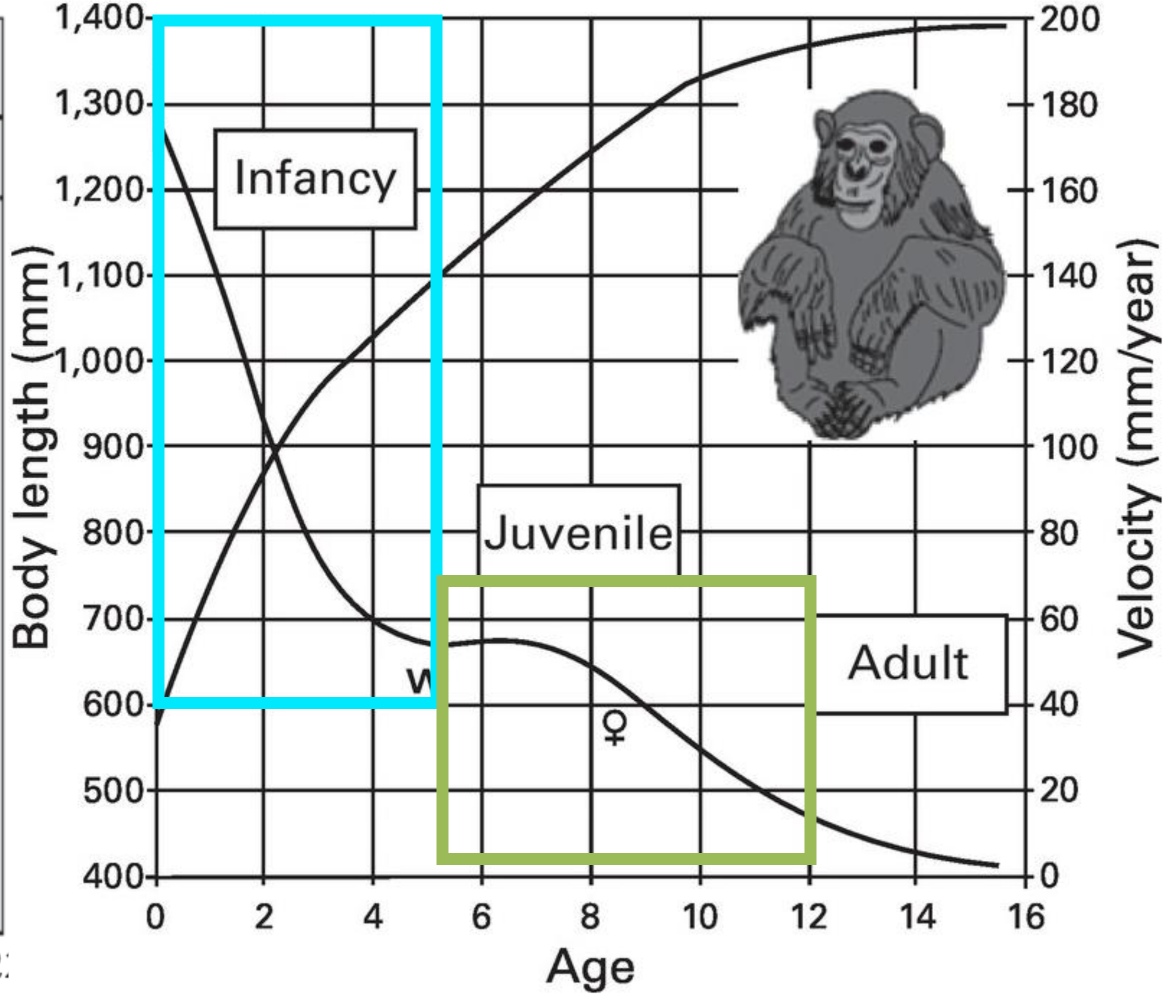

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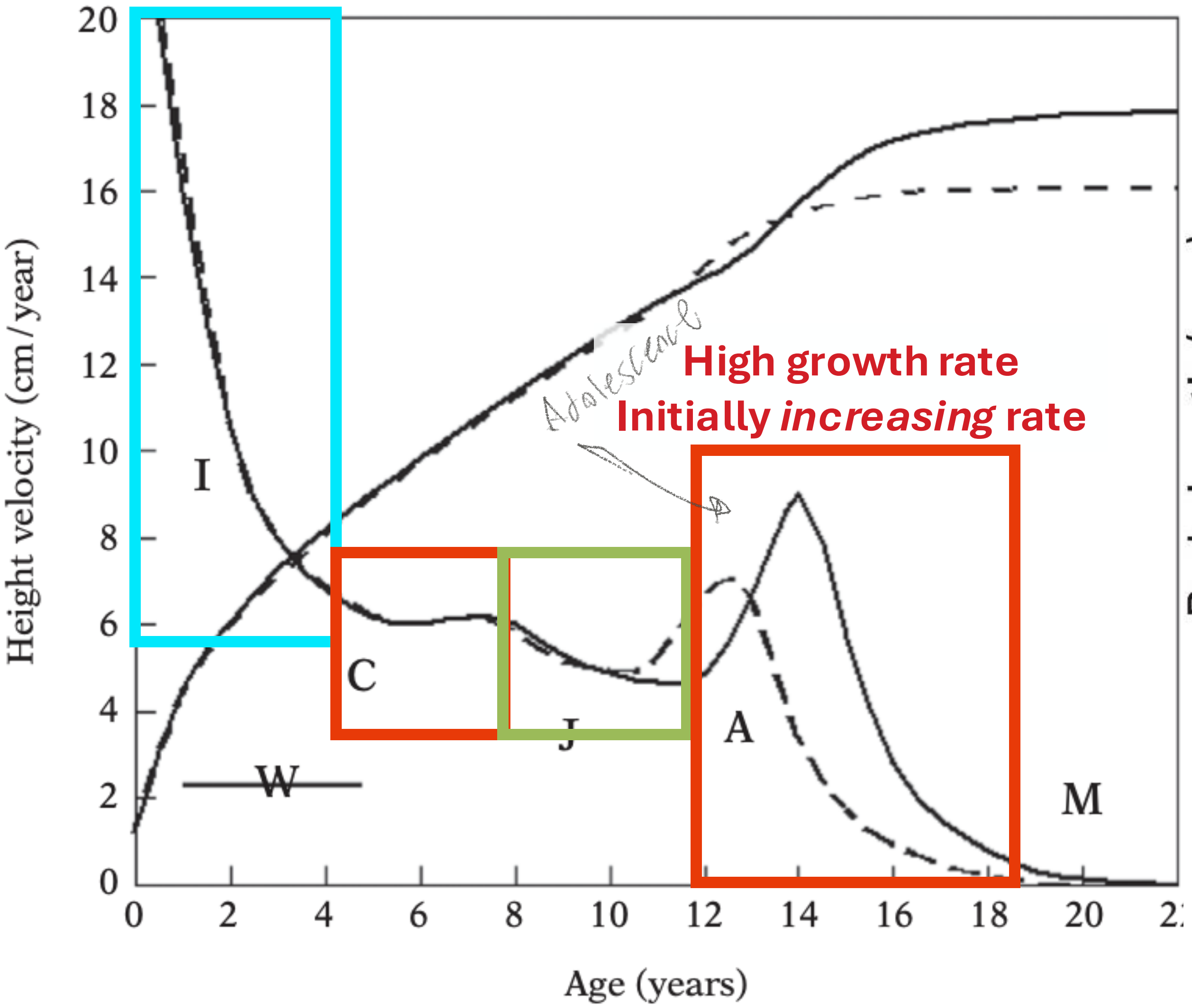
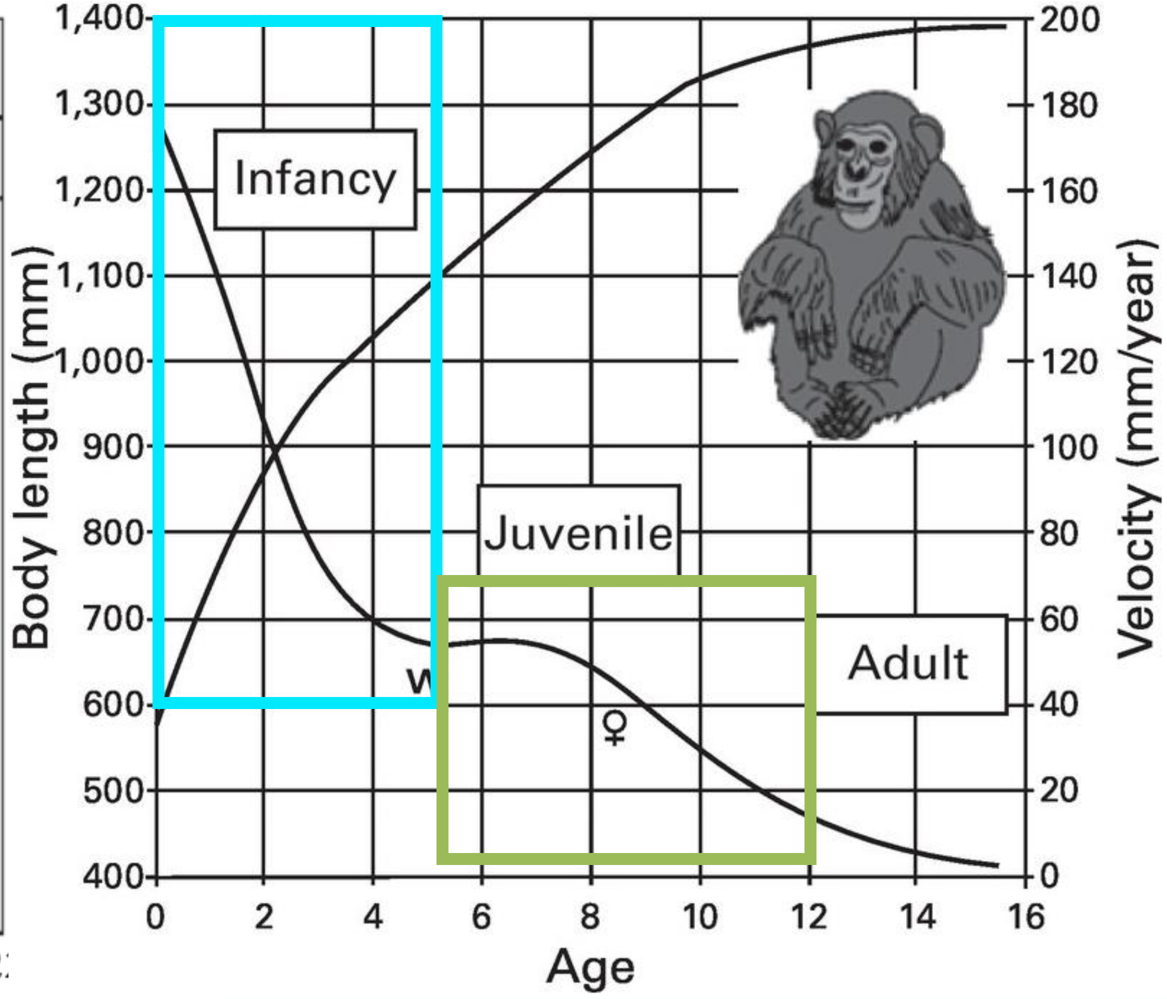
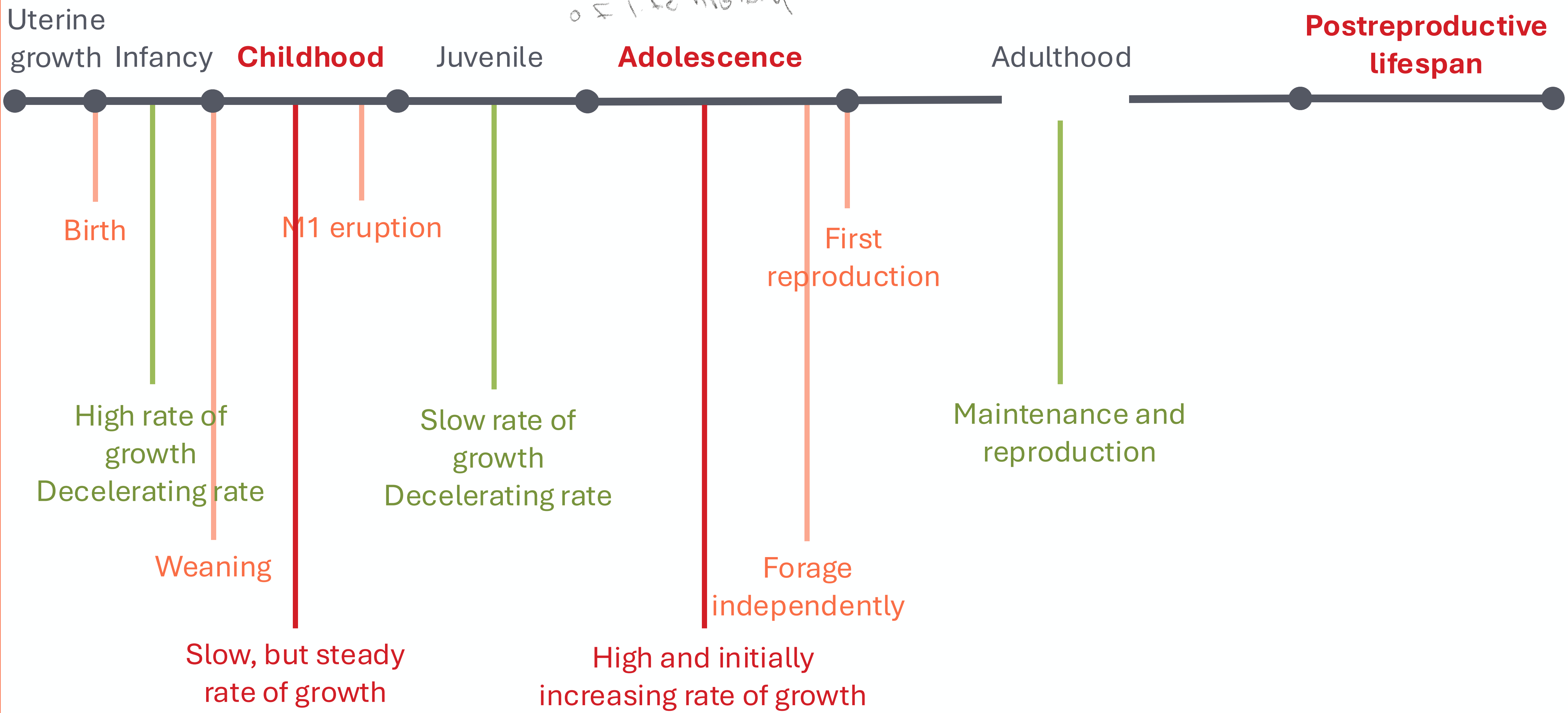




Key!!!

Human life history

Humans have 3 extra periods of life history



What?

Why?

When?

Why does human life history evolve?

What do we have to explain?

Childhood

- **Early weaning**
- **Steady rate of body growth**
- **High rate of brain growth**

focus for this lecture

patterns of human life history

Adolescence

- **High rate of growth**
- **Growth spurt**

Postreproductive life

- **Life is about turning energy into offspring...**

Generally longer & slower life

Why does human life history evolve?

What do we have to explain?

Childhood

- Early weaning
- Steady rate of body growth
- High rate of brain growth

Adolescence

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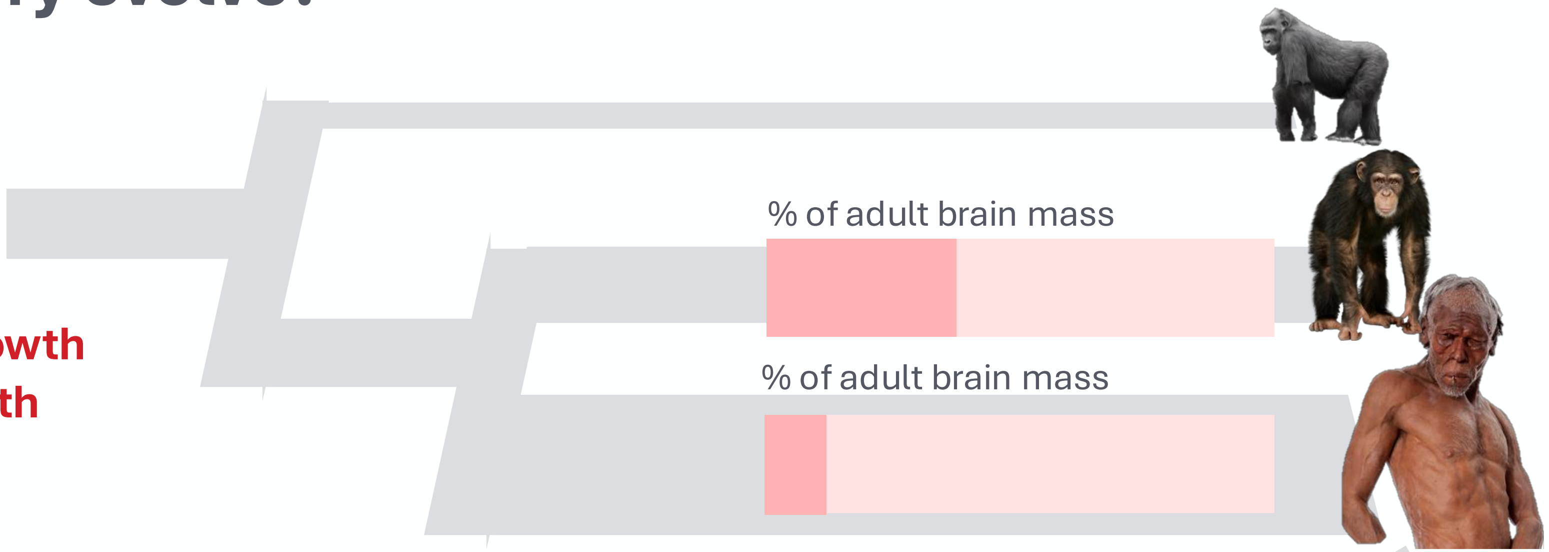


- Huge brains
- Born especially altricial: would need 21 months of gestation to be born at the equivalent stage to chimpanzees

But would make birth difficult

Why does human life history evolve?

What do we have to explain?

Childhood

- **Early weaning**
- **Steady rate of body growth**
- **High rate of brain growth**

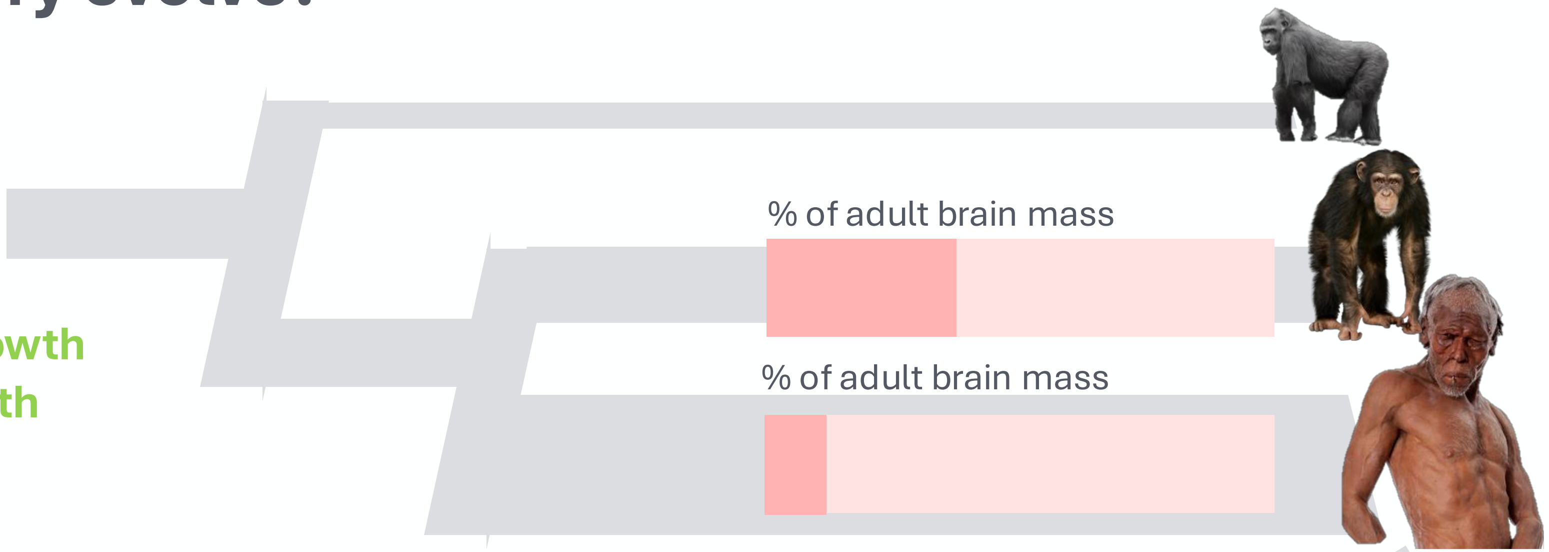
Adolescence

- **High rate of growth**
- **Growth spurt**

Postreproductive life

- **Life is about turning energy into offspring…**

Generally longer & slower life



- Huge brains
- Born especially altricial: would need 21 months of gestation to be born at the equivalent stage to chimpanzees

childhood compensates for altriciality

Need more time, during ontogeny, for brain growth] childhood devoted to brain growth
But costly – so body growth rate kept steady

What?
Why?
When?

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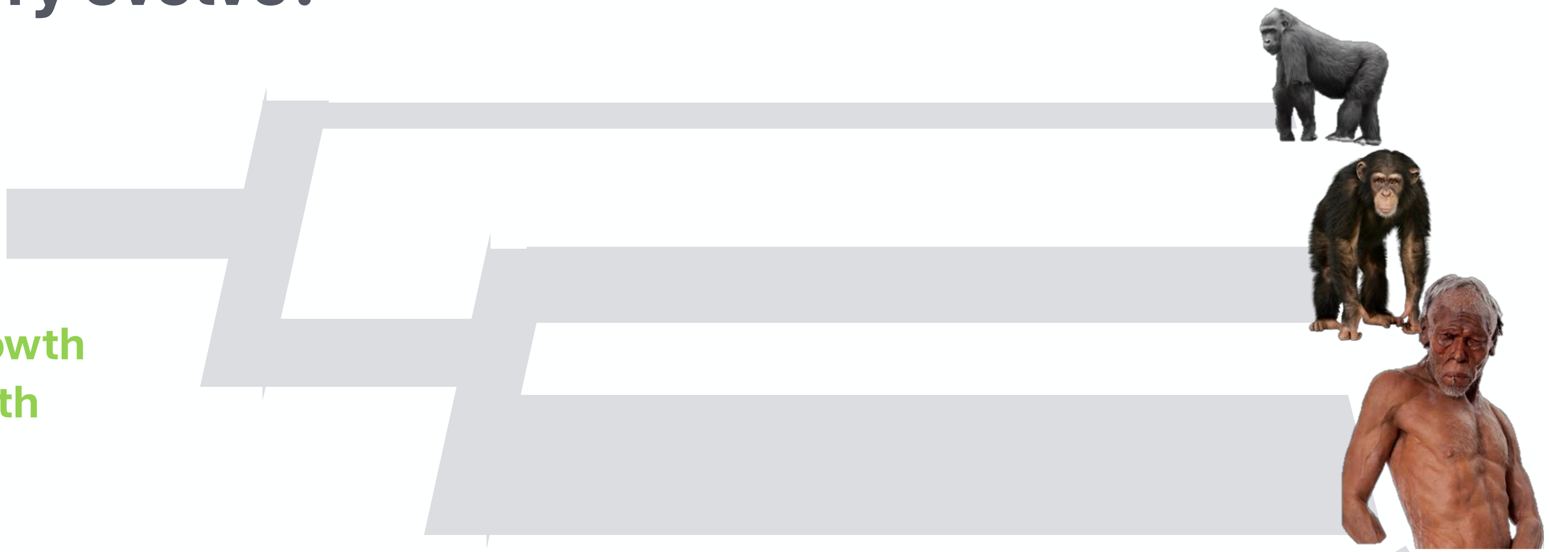
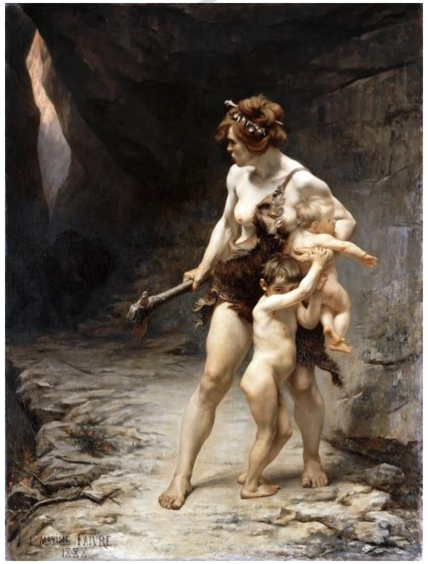


Early weaning allows human females to have multiple dependent offspring

Remember: evolution acts on **fitness** (lecture 3!)



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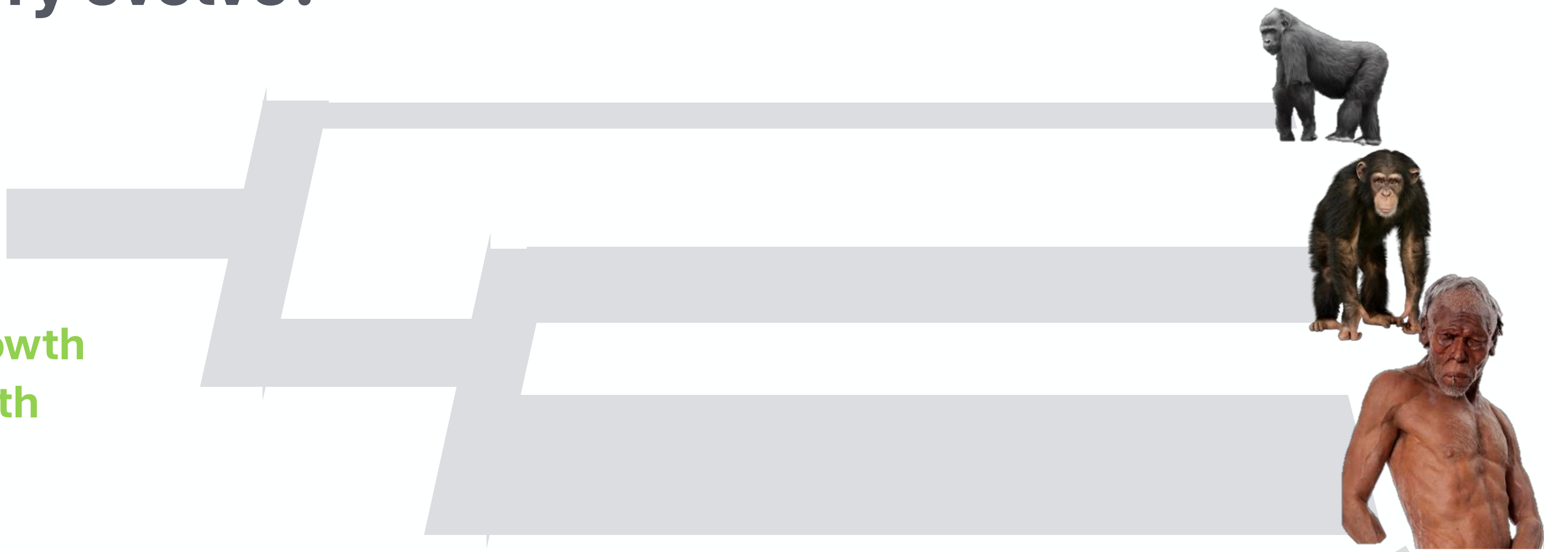
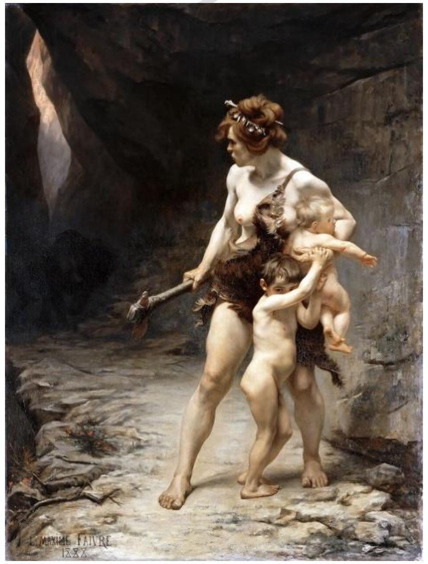


But wait…

energy



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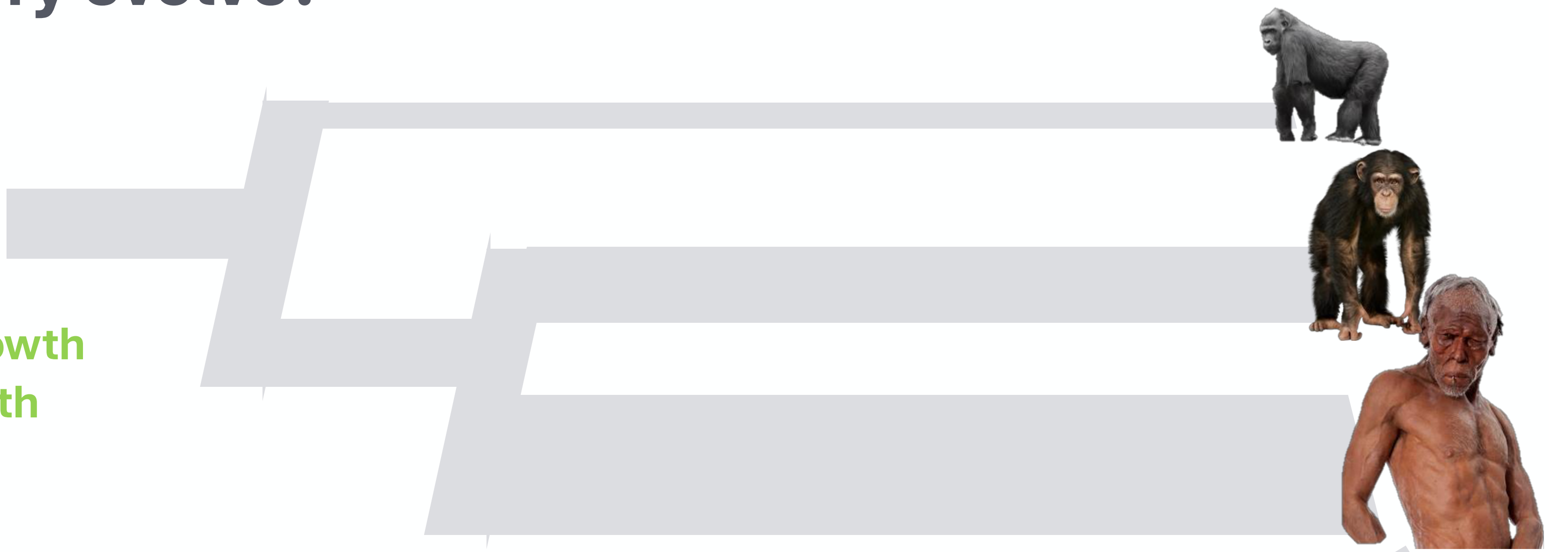

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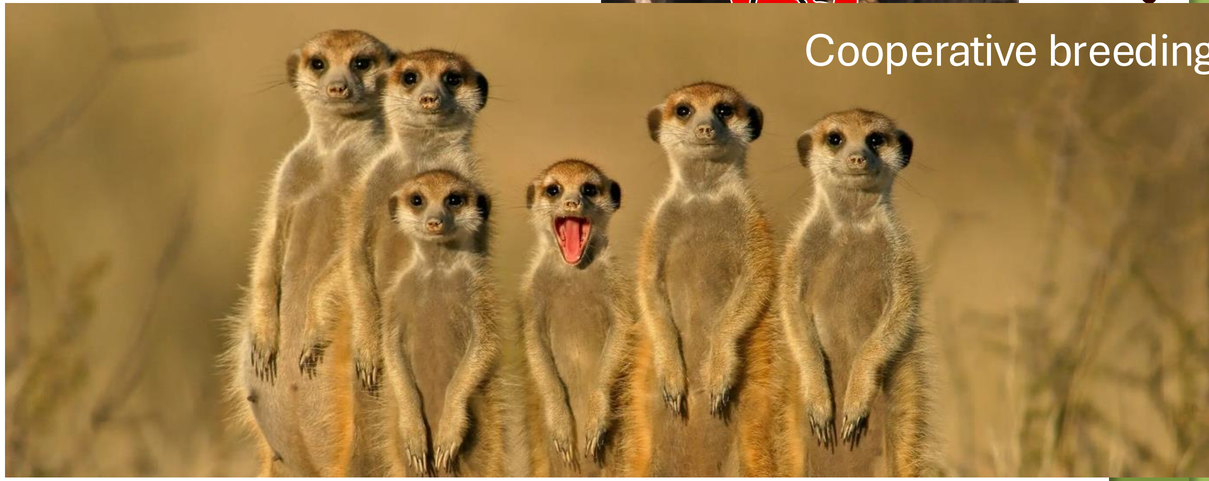

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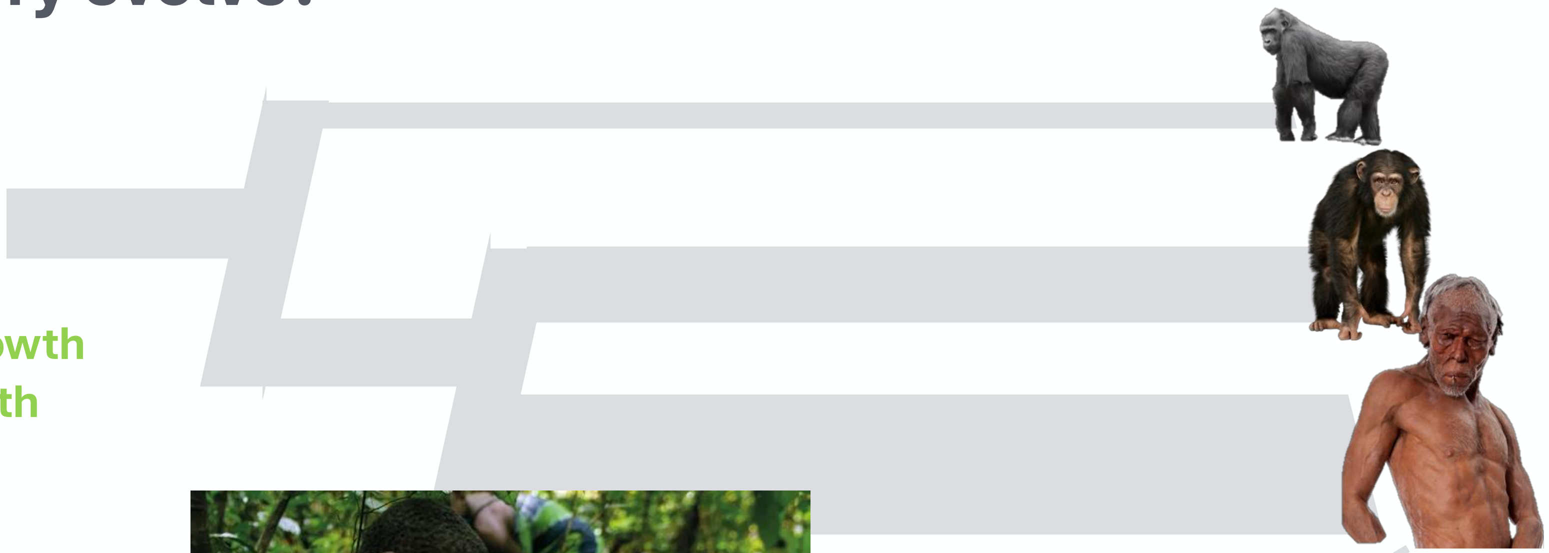

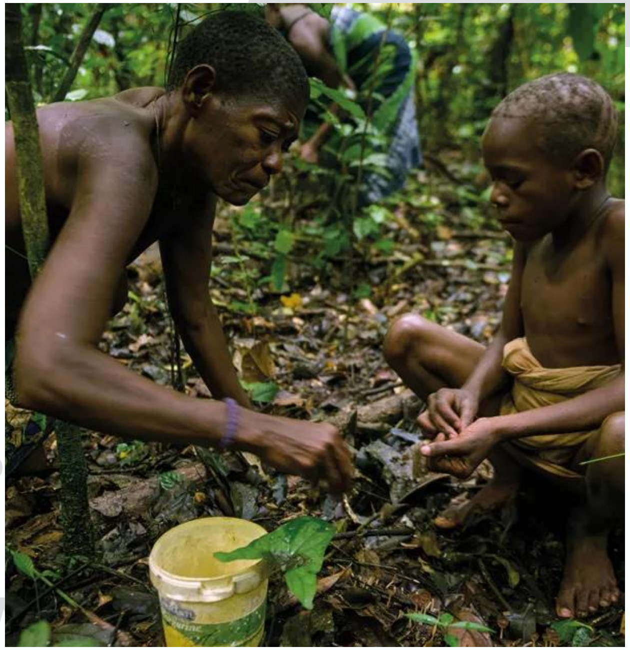

Childhood: important for learning

Why does human life history evolve?

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Key

- Extended period to allow brain to grow (necessary because larger brain *and* born with relatively smaller brain)
- Childhood allows mothers to offset own long growth period by having multiple dependent offspring
  - Cost is offset through alloparenting
  of multiple dependent offspring
- Childhood allows child to learn

Take-home message:

Evolution of childhood necessitates a very complex set of changes – to the individual but also the community it grows up in

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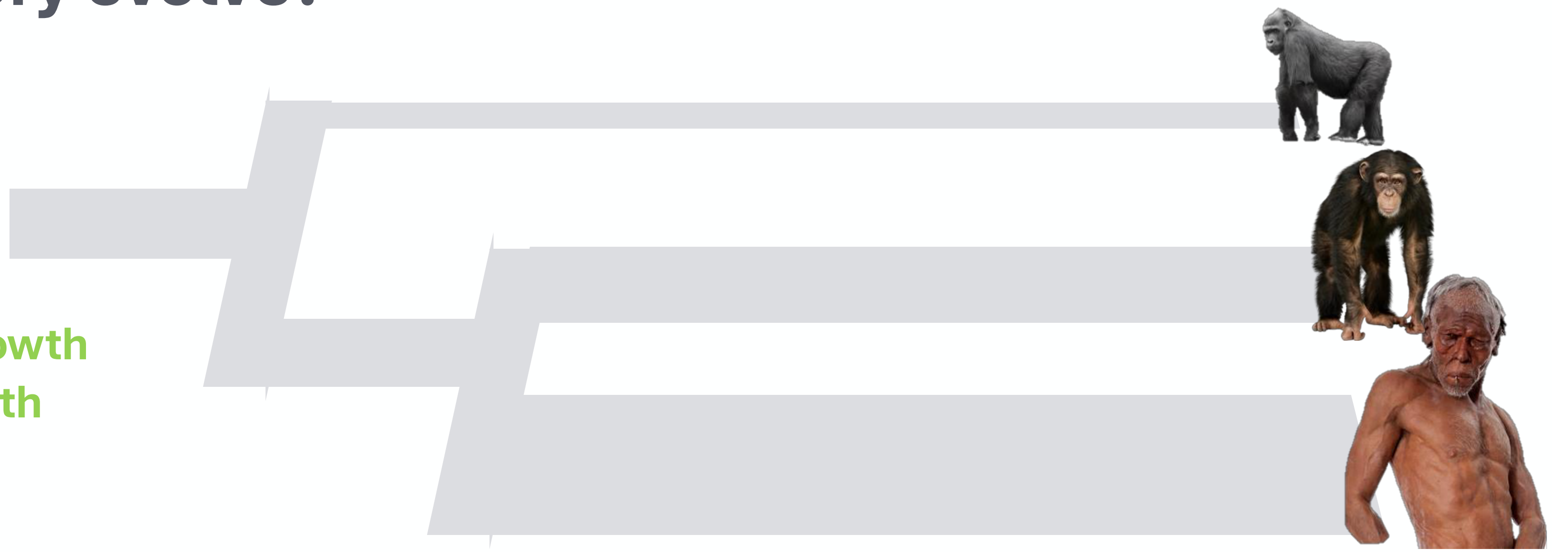

Adolescence

- High rate of growth
- Growth spurt

- Learning (but why that growth spurt)?
- Cooperative breeding (but why that growth spurt)?
- ‘Safe’ sexual practice (look mature but not as fertile as adults)? → But fertility levels are fairly high in adolescents . .
- Catch-up of skeletal growth: earlier phases marked by **brain** growth, not somatic growth

Postreproductive life

- Life is about turning energy into offspring…

Generally longer & slower life

Key

Non-mutually exclusive hypotheses for the evolution of adolescence – but these are hard to test

The big take-home messages

- What?
  - Life history is a species-specific pattern of energy allocation to biological functions
    - These functions can be roughly divided into those relating to growth, those relating to maintenance, and those relating to reproduction
  - Life history stages are characterized by distinct allocation patterns
  - General mammalian life history is somewhat tinkered with in primates (M1 erupts before weaning, long life); this primate pattern is completely broken in humans
  - Human life history has three unique stages: childhood, adolescence, and a postreproductive period
- Why?
  - Hard to say – many non-mutually exclusive explanations. However, likely very much due to our large brains!
- When? → moved to next lecture
  - ~~Between 1.8 and 0.5 million years ago~~